

**Human Resource Management: Performance Management
and Employee Performance Analytics.**

Presented to the Faculty of the
BESTLINK COLLEGE OF THE PHILIPPINES

In Partial Fulfilment
Of the Requirements for the Capstone:
Bachelor of Science in Information Technology

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MARCH 2024

DECLARATION

The researchers of this group certify that this project study does not incorporate, without acknowledgement, any material previously submitted for a degree of Diploma in any University and to the best of our knowledge and belief. It does not contain any material previously published or written by another person or ourselves except where due reference is made in the text. we also hereby give consent for our Project Study. If accepted, to be made available for photocopying and for interlibrary and for the title and summary to be available to outside organizations.

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DEDICATION

The researchers of the group wholeheartedly dedicate the humble peace of work to good and to our families, relatives, friends, colleagues and teachers for their consistent help and support that has been the source of our strength, inspiration and humility to pursue our endeavor.

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ABSTRACT

Title	HUMAN RESOURCE MANAGEMENT SYSTEM: PERFORMANCE MANAGEMENT SYSTEM AND EMPLOYEE PERFORMANCE ANALYTICS.
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CHAPTER 1

1. Introduction

A performance management system tracks the performance of employees in a manner that is consistent and measurable. The system relies on a combination of technologies and methodologies to ensure people across the organisation are aligned with – and contributing to – the strategic objectives of the business.

The system is collaborative, with managers and employees working together to set expectations, identify employee goals, define performance measurement, share employee performance reviews and appraisals, and provide feedback.

1.1 Background Of The Capstone Project

This project aims to create a system that can help them to computerized their manual HR Department. This project will result in the development of a new system and supports corporate strategy of providing progressive solutions to HR Department which improve productivity in the workplace.

1.2 Context And Scope

The Scope Management Plan provides the scope framework for this project. This plan documents the scope management approach; roles and responsibilities as they pertain to project scope; scope definition;

verification and control measures; scope change control; and the project's work breakdown structure. Any project communication which pertains to the project's scope should adhere to the Scope Management Plan.

This project is for designing, programming, and testing a new software product which will be used to computerized the manual HR department. This includes design of the software, all programming and coding, and testing/validation of the software. No external resources or outsourcing are anticipated for this project.

1.3 Problem Statement

Since its inception, HR dept. has relied upon a mainframe system to manage the performance of the employee and other sub-system functions. As the number of employees grows, so does the burden placed upon headquarters to effectively manage the company's administration at acceptable levels. In the past years HR dept has hired employees into overhead positions to help manage and run the day-to-day administration operations. These positions provide little or no return on investment as they are not billable positions and only maintain the status quo; they do nothing to improve the management of the company's administration. Additionally, employees must currently call their regional managers to enter their work hours and raise any concerns regarding employee performance

and administrative tasks. This places a large burden on managers who must balance these requirements with their day-to-day billable tasks.

Reporting is another problem area associated with the legacy mainframe system. All weekly and monthly employee performance reports must be generated manually which allows for a high probability of error and require significant amounts of time. These manual tasks further add to the burden and expense of the company.

1.4 Objectives And Goals

The Project directly supports several of the corporate goals and objectives. The following table lists the business goals and objectives that the Project supports and how it supports them:

Business Goal/Objective	Description
Timely and accurate reporting	Web based tool will allow real-time and accurate reporting of all the information needed from other sub-system
Improve staff efficiency	Fewer HR staff required for managing these activities will improve efficiency

TABLE 1: GOALS AND OBJECTIVES

1.5 Significance And Relevance

This project will result in greater efficiency of day-to-day employee performance and analytics, significantly lower overhead costs, and reduced turnover as a result of providing employees with greater autonomy and flexibility. Additionally, managers will once again be focused on billable tasks instead of utilizing a significant portion of their time on non-billable administrative tasks.

1.6 Structure Of The Document

This document is organized into several sections, each focusing on a specific aspect of the capstone project. The subsequent sections will delve into the project's technical and architectural details, development process, testing, and quality assurance, as well as the results, evaluation, and lessons learned. By the end of this document, readers will gain a comprehensive understanding of our sub-system and its impact on the HR's operational efficiency and effectiveness.

2. Literature Review

2.1 Agile Scrum Methodology Overview

1. A company's culture not being aligned with the value of the agile scrum approach is often seen as the most significant challenge to effectively implementing the agile approach (Cappelli & Tavis, 2018).
2. Also in regards to compensation, organization need to focus on group level pay for performance policies not just individual level as agile scrum employees are producing a product or service that requires an interdependent group (Darrell, et. al., 2016).
3. Specifically companies need to view career advancement as developing an agile scrum members competencies so they can expand their influence and increase their recognition(financial and non – financial) all while increase their value to an organization (Rigby, et. Al., 2018).
4. Therefore, a more agile approach to human resources in general is likely needed in the future (Cappelli & Tavis, 2018).
5. It is possible to change over significant organizational operations to an agile approach all at one time like ING did in 2015 (Barton, et. Al., 2018)

2.2 Enterprise Architecture Concepts

1. Marc Lankhorst (Lankhorst et al. 2011) defines enterprise as:
Enterprise architecture is a coherent whole of principle, methods, and models that are used in the design and realization of an enterprise's organizational structure.
2. Enterprise architecture can also be viewed as a master plan facilitating the convergence of diverging forces towards a strategic transformation target, like a targeted business model or a targeted operating model (Schekkerman 2004).
3. John Zachman (1987) John Zachman (1987) applied the notion of architecture to represent the functioning of an enterprise, specifically within the context of the enterprise IT landscape and the business environment.
4. Paul Estrach (2023) Enterprise architecture is a framework that helps businesses align their strategy, processes, and technology to achieve their goals.

2.3 Relevant Studies And Research

1. Particularly, research suggests investigating PM's effects on workers' motivation, job engagement, productivity, and retention (Audenaert et al., 2019).
2. Because developing countries' organizational structures, cultures, and performance differ from those of the developed world, it is crucial to recognize that these differences may have varying consequences for PM (Audenaert et al., 2019).
3. Studies indicate that the management of employee performance can be achieved by effective control over its determinants, and as such, this approach should be investigated (Kang & Choi, 2019).
4. The key components of PMSE have been identified as effective goal setting, control and feedback, measurement, alignment with objectives, performance-based rewards, and a favorable view of fairness (Sharma et al., 2016).
5. Performance at the individual level improves organizational performance, according to Zhong et al. (2016).

2.4 Integration Of Information Systems In Enterprise Environments:

Human resource management, or HRM, has seen substantial changes in recent years. The advancement of information technologies in this sector of knowledge has greatly aided in the shift in focus from administrative management responsibilities to becoming a strategic partner of the overall organization plan. The manner that HRM is administered nowadays is profoundly impacted by the widespread use of information systems. It accelerated a significant change in the methods and procedures used by organizations for human resources (HR), specifically with regard to the gathering, storing, using, and sharing of data. Numerous HRM procedures have improved in efficiency, and the effect of this service level increase has made it possible for HR to play a bigger role in the business plan.

3. Methodology

3.1 Agile Scrum Method In Projects

The Project Manager, Research Adviser and team will all play key roles in managing the scope of this project. As such, the research adviser, project manager, and team members must be aware of their responsibilities in order to ensure that work performed on the project is within the established scope throughout the entire duration of the project.

The table below defines the roles and responsibilities for the scope management of this project.

3.2 Roles:

1. Scrum Master: The Scrum Master, accountable for facilitating Scrum events, removing obstacles, and making sure the Scrum process is followed.
2. Product Owner: The Product Owner is the primary project stakeholder, articulating specifications, setting feature priorities, and representing the interests of the HRM department.
3. Development Team: The actual development work is carried out by a cross-functional development team made up of people with a variety of backgrounds and specializations.

3.3 Sprint Cycles

Regular check-ins and progress evaluations are part of sprint cycles. This could entail holding more frequent feedback meetings between managers and staff in performance management. These check-ins offer the chance to talk about accomplishments, difficulties, and prospective goal revisions.

3.4 Scrum Artifacts

1. Product Backlog: A product backlog in performance management could be an employee's prioritized list of skills, competencies, or areas for development. It is possible to periodically review and modify this backlog.
2. Sprint Backlog: A sprint backlog in performance management could be an employee's commitment to completing particular tasks or goals in a shorter amount of time.

Product Backlog (User Stories)

User Story No.	USER STORIES	User Story Priority	Revision Priority	Status
1.	As an HR ADMIN, I can log in accountable System. <i>Can log in</i>	1	5	done
2.	As an HR ADMIN, I analyse the absentees of the employees <i>Can Analyze</i>	1	5	done
3.	As R Admin, I can view activity logs <i>Can view activity</i>	1	4	done
7.	As an HR ADMIN, I can track employee work hours and overtime	1	5	done

8.	As an HR ADMIN, I can set up various pay structures including hourly wages and salaries	1	4	done
9.	As an HR ADMIN, I can update personal information.	1	4	done
10.	As an HR ADMIN, I will provide security and data protection.	1	4	done
11.	As an HR ADMIN, I can log out.	1	5	done
12.	As an Employee, I can log in using OTP through E-Mail	1	5	done
13.	As an Employee, I can view the dashboard.	1	4	done
14.	As an Employee, I can view attendance	1	3	done
15.	As an Employee, I can request leave	1	5	done
16.	As an Employee, I can request certificate	1	4	done

17.	As an Employee, I can view and export payslip	1	5	done
18.	As an Employee, I can create appointments	1	5	done
19.	As an Employee, I can view Articles and Programs	1	4	done

Table 2: Product Backlog

Product Backlog for EIS Information Security

User Story No.	USER STORIES	User Story Priority	Revision Priority	Status
20.	As a Security Personnel, I need to be able to enter my username and password to access the system.	1	5	done
21.	As a Security Personnel, I want to put two factor authentication to avoid hackers.	1	5	done

Table 3: Product Backlog for EIS Information Security

Product Backlog for EIS Standards

UI/UX

User Story No.	USER STORIES	User Story Priority	Revision Priority	Status
22.	As an Owner, I want the system's logo to appear.	1	4	done
23.	As an Owner, I want functionally clear icons that are clear and consistent throughout the system.	1	5	done
24.	As an Owner, I want the font to be medium size so that clients can read it.	1	5	done
25.	As an Owner, I want design a system that has a	1	4	done

	week unified color palette and has a pleasing user interface.			
26.	As an Owner, I desire a responsive design for the system, giving a consistent user experience across all devices.	1	5	done
27.	As an Owner, I want the interface to be simple to use with obvious choices and button placement.	1	4	done

Table 4: Product Backlog for EIS Standards

Product Backlog for Integration.

User Story No.	USER STORIES	User Story Priority	Revision Priority	Status
28.	As a HR ADMIN, I want the human resources performance management to be integrated to other sub modules.	1	5	done
29.	As a HR ADMIN, I want the human resource employee performance analytics to be integrated to other sub modules.	1	5	done

Table 5: Product Backlog for Integration.

3.5 Integration Approach For Information Systems

Clearly state what the integration initiative's goals are. Determine which particular performance management data and processes require integration. To make sure the integration satisfies users' and stakeholders' needs, be aware of their requirements.

3.6 Introduction To Togaf And The Four Architectural Domains

The purpose of TOGAF is to help businesses create an information systems architecture that meets their needs. Enterprise architecture is divided by TOGAF into four domains, each of which focuses on a particular facet of the company:

1. Business Process Architecture: focuses on comprehending and recording the organization, governance, organization structure, and important business processes.
2. Application Architecture: focuses on the relationships and interactions between software applications and how they are designed and organized.
3. Data Architecture: describes the logical and physical data assets of an organization, their management, and their storage.

4. Technology Architecture: focuses on the network infrastructure, hardware, and software needed to enable the deployment of applications.

Although TOGAF was not created with performance management in mind, it does offer a strong framework for coordinating business and IT strategies. Organizations can make sure that their performance management initiatives are strategically aligned with overarching business goals by addressing the domain of business architecture. Furthermore, data architecture can help to guarantee that performance-related data is managed effectively.

To put it briefly, TOGAF is a thorough framework for enterprise architecture, and its four architectural domains offer an organized method for handling various organizational aspects. By using TOGAF principles, organizations can ensure a strategic and all-encompassing approach to managing enterprise architecture and better align performance management initiatives with overarching business objectives.

4. Requirements Analysis

4.1 Stakeholder Identification

The first stage of requirements analysis is to identify the important stakeholders. Stakeholders in our HRM system project consist of:

1. Leadership and Management: senior executives and leaders who establish the overarching objectives and plans of the organization.
2. Employees: workers of different ranks who directly aid in the accomplishment of company objectives.
3. Human Resources: Participates in process design, implementation, and management for performance management.
4. Project Team: those or groups in charge of putting new performance management systems or upgrades into place.

4.2 Requirements Gathering Techniques

To encompass the various demands of stakeholders, we utilize an assortment of requirements gathering methodologies:

1. Surveys: Send surveys to managers, staff members, and other relevant parties to get their opinions on the performance management procedures in place as well as their suggestions for enhancements.

2. Interviews: To learn more about the opinions of key stakeholders regarding the current performance management system and what they would like to see improved, conduct interviews with managers, HR specialists, and workers.
3. Workshops: Conduct workshops where stakeholders can actively participate in brainstorming sessions to generate ideas for improving performance management processes. This can include defining key performance indicators (KPIs) and setting goals collaboratively.
4. Observation: To gain insight into the current management of performance expectations, watch managers and employees at work. Determine your areas of pain and improvement.
5. Document Analysis: Analyze documents such as current performance appraisal forms, job descriptions, and organizational policies related to performance management. This helps in understanding existing structures and processes.

4.3 User Stories And Use Cases

A user story is a brief, straightforward account of a feature or functionality written from the viewpoint of the end user. A use case is a thorough explanation of how a system communicates with an outside party (a user or another system) in order to achieve a particular objective.

4.4 Functional Requirements For Integration

Functional requirements are defined to outline the specific functionalities of the HRM system. These requirements include:

1. **User Authentication and Authorization:** To protect the confidentiality and integrity of data, it is necessary to guarantee secure user authentication and authorization across integrated systems.
2. **Real-time Reporting and Analytics:** Streamline reporting and analytics in real-time by combining performance data from multiple sources.
3. **Integration with Learning and Development Systems:** To match employee training and skill development with performance management, integrate with learning and development systems.
4. **User Training and Support:** Provide training and support resources for users to understand and leverage the integrated performance management systems effectively.

5. Business Process Architecture

5.1 Identification of Business Process Architecture

Identification of Business Processes: Begin by identifying and defining the key business processes within the organization or domain relevant to your project. These processes are the fundamental steps and activities that drive the organization's operations.

It's crucial to work closely with stakeholders to ensure an accurate understanding of these processes. Discuss the significance of identifying these processes of the project and how they impact the organization's success.

5.2 Business Process Diagrams

Create and present business process diagrams that visually represent the identified processes. Use recognized notation such as Business Process Model and Notation (BPMN) to illustrate the flows, tasks, decisions, and interactions within these processes. These diagrams provide a clear and visual representation of how the business operates, making it easier for stakeholders to understand and analyze

BUSINESS PROCESS ARCHITECTURE LEVEL 1:

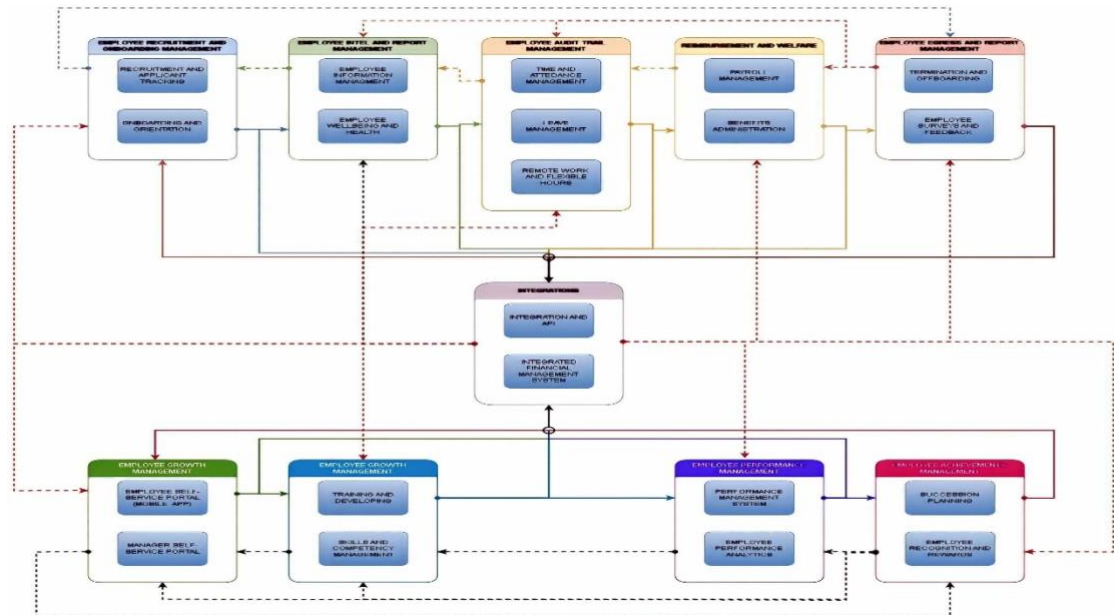


Figure 1. BPA 1

BUSINESS PROCESS ARCHITECTURE LEVEL 2:

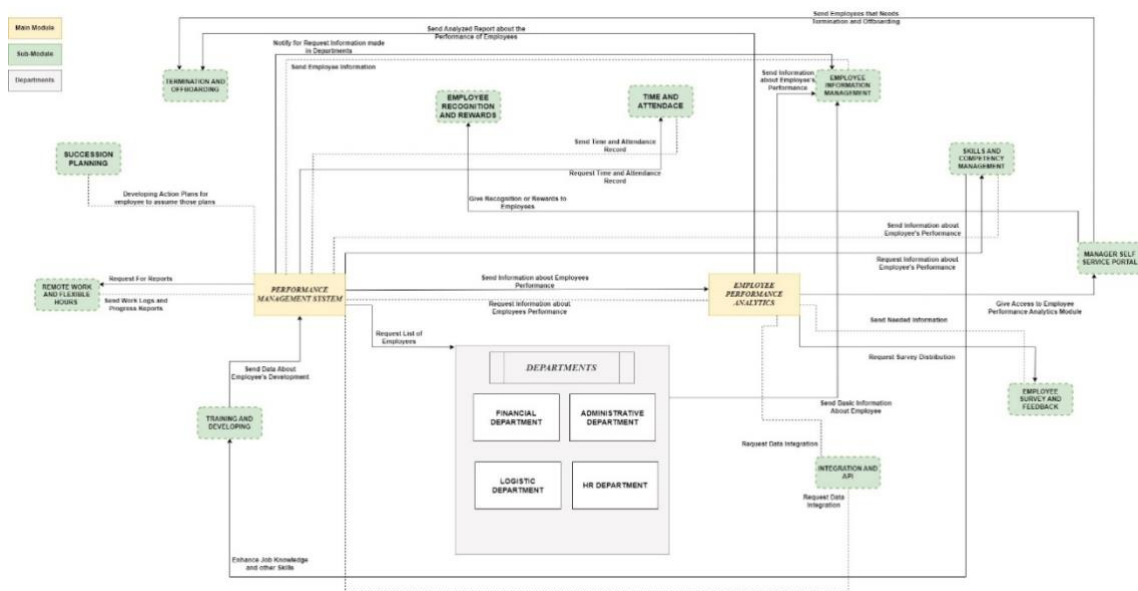


Figure 2. BPA 2

5.3 Alignment Of Integrated Systems With Business Processes

Explain how integrated systems align with an organization's business processes. Detail the specific integration points where the system interacts with these processes. Describe how integrated systems support, improve, or automate aspects of business processes. Discuss how this alignment benefits the organization, such as improving efficiency, reducing errors, or enabling new features.

5.4 Business Process Improvement

Discusses how integrated systems have the ability to deliver improvements to an organization's business processes. Highlight any identified opportunities to streamline processes, reduce manual effort, or improve overall workflow. Explain how these improvements contribute to project goals and organizational goals. If possible, provide before and after scenarios to illustrate the impact of these improvements.

6. Application Architecture

6.1 Components Of Application Architecture

The application architecture of our system comprises several key components, each serving a specific purpose in the overall functionality:

1. **User Interface (UI):** An intuitive user interface that gives managers and staff access to a dashboard where they can view and track data linked to performance. a specialized interface for managers and HR specialists to examine staff management data.
2. **Business Logic:** This procedure is essential for enhancing employee productivity and engagement, matching team and individual performance to corporate objectives, and making well-informed decisions on talent development and retention. The essential components of the business logic are Goal Alignment, Performance Measurement, Employee Development, etc.
3. **Database Management System (DBMS):** Employee data, including personal information, work history, skill sets, and other pertinent information, is stored in a database managed system (DBMS). Each employee's goals, performance metrics, and key performance indicators (KPIs) are kept in a database.
4. **Application Programming Interface (API):** In conclusion, APIs improve employee performance analytics and HR performance

management by enabling data transfer, integration, and communication across various software programs. As a result, the HR ecosystem becomes more integrated and effective, facilitating well-informed decision-making and enhancing overall organizational performance.

6.2 Application Architecture Diagram

To visually represent the application architecture, we create architecture diagrams illustrating the components' relationships and interactions. These diagrams help in understanding how data flows, how user interactions are processed, and how different modules communicate with each other.

APPLICATION ARCHITECTURE DIAGRAM:

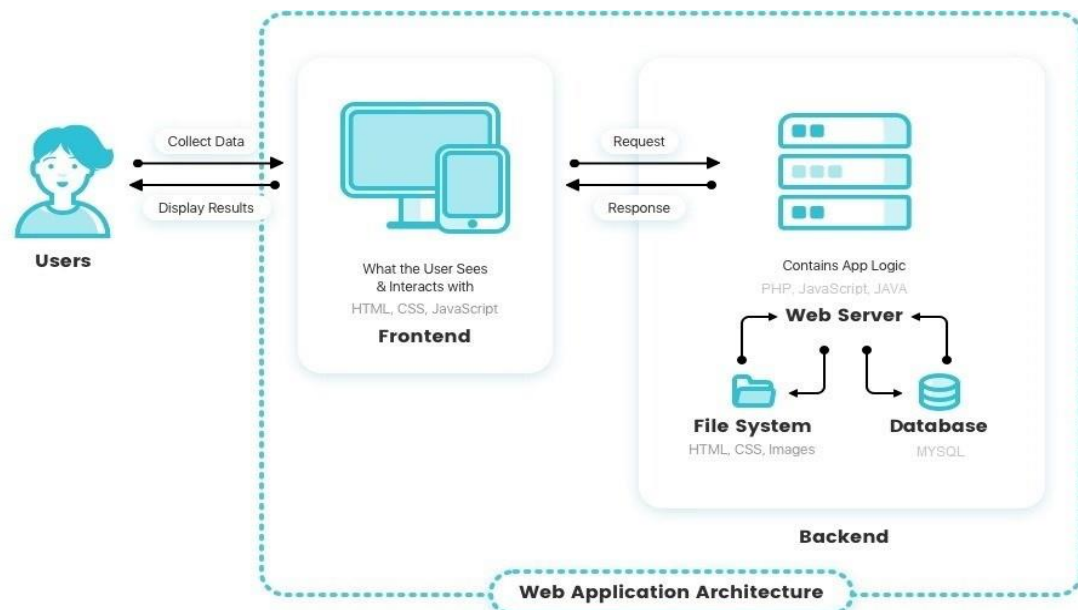


Figure 3. Web Application Architecture Diagram

6.3 Integration Of Software Modules

In a HRM system, various software modules need to work together seamlessly. Integration is achieved through the use of APIs and middleware. Setting performance objectives, managing employee performance generally, and performing performance reviews fall within the purview of this module. In order to identify important patterns and insights, this module analyzes employee performance data.

6.4 Communication And Interaction Patterns

Efficient communication and interaction patterns are essential to the application's performance and user experience. In our application architecture, we adopt the following patterns:

1. **CLEAR COMMUNICATION OF PERFORMANCE EXPECTATIONS:** HRM should make performance objectives crystal apparent to staff members. Setting SMART (specific, measurable, attainable, relevant, and time-bound) objectives is necessary for this. Employees are better able to comprehend their duties and responsibilities when performance goals are communicated on a regular basis, which promotes responsibility.

2. TRAINING AND DEVELOPMENT OPPORTUNITIES: Employee development depends on open communication regarding opportunities for training and development. Employees should be made aware of the programs and tools that HRM offers to help them develop their skills. The organization's overarching goals should be in line with training discussions and connected to performance targets.
3. RECOGNITION AND REWARDS: Positive conduct is reinforced through communication regarding incentives and recognition. The standards for incentives and the connection between achievement and acknowledgment should be made explicit by HRM. When used appropriately, public acknowledgment may enhance a pleasant work environment.

7. Data Architecture

7.1 Data Sources And Types

Begin by identifying and describing the data sources used in the integrated system. These sources may include databases, external APIs, file systems, or other data stores. Detail the types of data they store, such as structured (relational databases), semi-structured (JSON or XML), or unstructured (documents, images). Explain the importance of identifying data sources for the project.

7.2 Data Flow Diagrams

Create data flow diagrams that visually represent how data moves through an integrated system. Use standardized notation such as data flow diagrams (DFDs) or other appropriate visualization techniques. These diagrams provide an overview of how data is processed, transformed, and exchanged between different components of the system.

DATA ARCHITECTURE:

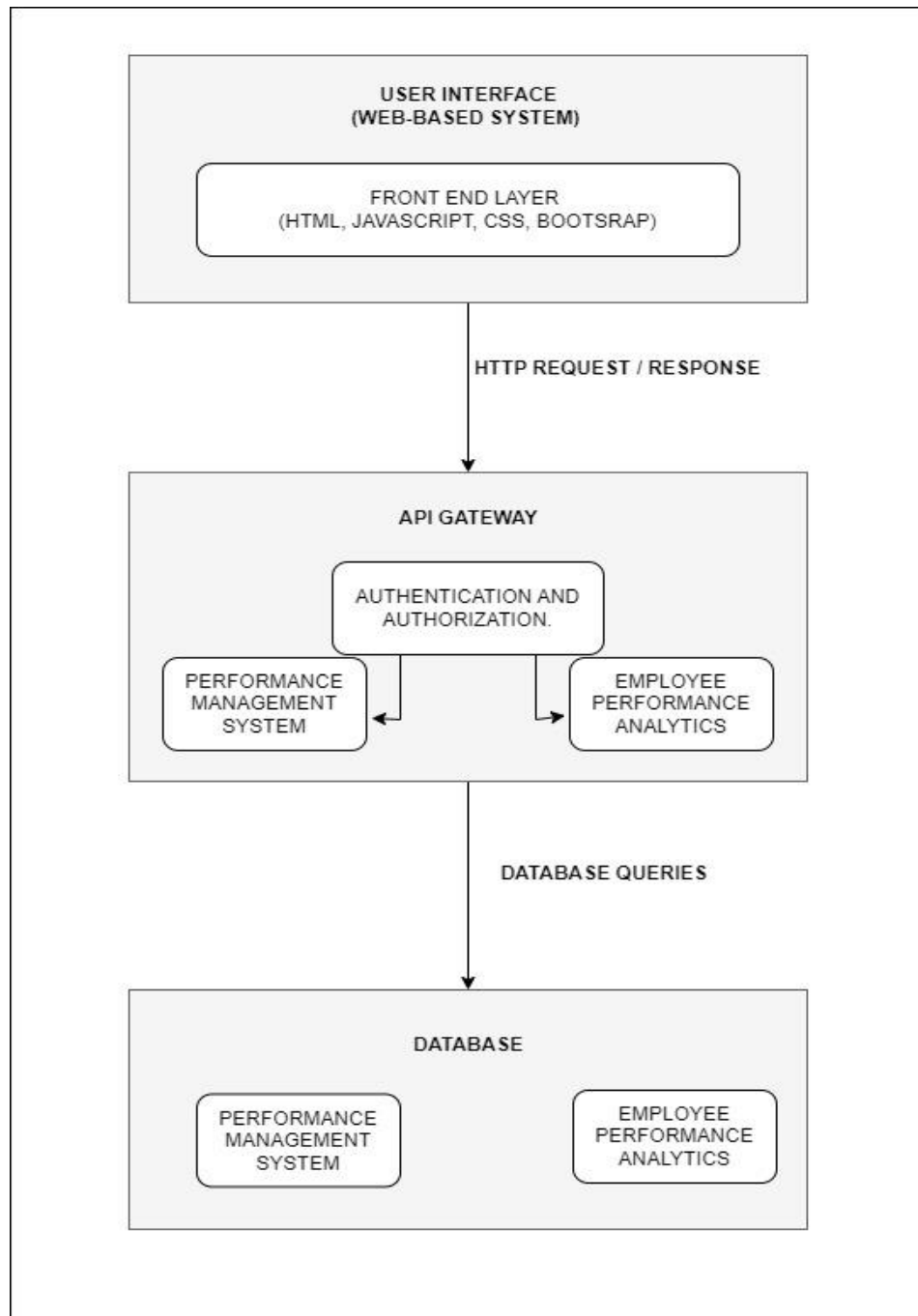


Figure 4. Data Architecture.

7.3 Data Storage And Management

Describes the architecture and methods used to store and manage data in an integrated system.

Discuss the database management system (DBMS) or data storage technology used. Explain how data is organized, indexed, and secured. Detailed data storage and retrieval procedures, including CRUD (Create, Read, Update, Delete) operations.

Synchronizing data between systems Explains how data is synchronized between integrated systems to maintain data consistency and integrity. Discusses the mechanisms, protocols, and tools used for data synchronization. Addresses all applicable data mapping and transformation processes to ensure compatibility between systems. Emphasize the importance of synchronization for seamless data sharing and decision making.

The “Data Architecture” section aims to define and illustrate how data is managed, stored, and synchronized in the integrated system. Accurate documentation of data sources, types, data flows, storage and synchronization are essential to ensure the quality, consistency and accessibility of data that is essential for success of project.

7.4 Data Synchronization Across Systems

Maintaining data consistency throughout the many modules of our system depends on data synchronization. We have put in place a data integration layer that synchronizes data changes across the different databases and systems in order to do this. Real-time data synchronization makes sure that every module has access to the most recent data. When real-time synchronization is not possible, data is updated on a regular basis using batch procedures that are planned.

Processes for data mapping and transformation are used to guarantee data consistency and interoperability across systems. This involves mapping data from one source to another, standardizing data formats, and resolving disputes.

In conclusion, the significance of data sources and types in the system is emphasized in our data architecture section. Data flow diagrams are used to illustrate data.

8. Technology Architecture

8.1 Technology Stack And Infrastructure

The technology architecture of our HRM system is built on a well-defined technology stack and infrastructure. We utilized a combination of open-source and industry-standard technologies to ensure scalability, security, and performance. Our technology stack includes:

1. **Operating System:** We deployed the system on Linux-based servers to ensure stability and security.
2. **Programming Languages:** Python was chosen as the primary programming language for backend development, while JavaScript was used for frontend development.
3. **Frameworks:** We leveraged the Django web framework for building the backend, ensuring rapid development, security, and modularity. React.js was chosen for the front end, providing a dynamic and responsive user interface.
4. **Database Management:** PostgreSQL, a powerful and open source relational database, is used to store and manage data.
5. **Web Server:** Gunicorn and Nginx are used to serve the application and handle HTTP requests.

6. Cloud Infrastructure: The system is hosted on Amazon Web Services (AWS) to ensure scalability, reliability, and ease of maintenance. AWS services, such as Amazon EC2 and Amazon RDS instances, are used to provide powerful database and storage capabilities.

8.2 Network Topology And Configuration

Our network topology followed a client-server architecture, where clients (users) interacted with the system hosted on the server.

Network configuration included:

1. Secure Socket Layer (SSL) encryption to secure data transmission.
2. Load balancing to distribute traffic across multiple server instances for improved performance and fault tolerance.
3. Firewall rules to control network access and protect against security threats.

8.3 Software Technologies

We employed various software technologies to enhance the functionality and user experience of the HRM system:

1. Django REST framework: This powerful framework was used for building robust APIs that allowed seamless data interaction between the frontend and backend.
2. React.js: A popular JavaScript library, React.js has made creating dynamic and responsive user interfaces easy.
3. Redux: This state management library is used to manage complex state and data flows in the application.
4. Docker: containers are used to isolate and deploy applications, ensuring consistency between development and production environments.
5. Jenkins: This CI/CD tool is used for automated testing, builds, and deployments, streamlining our development and deployment processes.

8.4 Scalability And Performance Considerations

To ensure scalability and performance of the HRM system, we have implemented the following considerations:

1. Dimensional scaling horizontal: The system is designed to be horizontally scalable, allowing us to add more server instances to accommodate increased user numbers.
2. Caching Mechanism: We have implemented caching to reduce database load and improve response time.
3. Database indexing: To optimize database queries, we have created appropriate indexes.
4. Performance Monitoring and Testing: Continuous performance monitoring and testing allows us to identify and resolve performance bottlenecks and ensure optimal system performance.

This section provides insight into the technological architecture that underpins our HRM system. It describes the technology stack, network configuration, software technology, and scalability and performance considerations. Our choice of technologies and infrastructure was carefully selected to support the system's robustness, security, and scalability.

9. Development Process

9.1 Agile Scrum Roles And Responsibilities

Name	Role	Responsibilities
Nicole Bueza	Project Manager	• Measure and verify project scope • Facilitate scope change requests • Facilitate impact assessments of scope change requests • Organize and facilitate scheduled change control meetings • Communicate outcomes of scope change requests • Update project documents upon approval of all scope changes
Angela Nicole L. Palomo	Team Lead	• Participate in impact assessments of scope change requests • Communicate outcomes of scope change requests to team • Facilitate team level change review process

Johnlester E. Francisco	Lead Programmer	• Evaluate need for scope change requests • Accept project deliverables
Justine Alen A. Sedano	Team Member	• Participate in defining change resolutions • Evaluate the need for scope changes and communicate them to the project manager as necessary
Dennis S. Saguisa	Team Member	• Participate in defining change resolutions • Evaluate the need for scope changes and communicate them to the project manager as necessary
Arvin-Jay O. Pajulas	Team Member	• Participate in defining change resolutions • Evaluate the need for scope changes

Table 6: Roles And Responsibilities

9.2 Sprint Planning And Backlog Management

Agile principles are being incorporated into performance management procedures by certain organizations. This entails establishing shorter-term performance targets and periodically assessing and modifying them in light of shifting organizational priorities. In Agile, sprint planning entails choosing a group of tasks to be finished in a predetermined amount of time.

9.3 Sprint Execution And Deliverables

During the course of development, we faced several obstacles. The project team will hold brainstorming sessions to determine the project's stakeholders. The principal project team and project sponsor will participate in the brainstorming session. There will be two sections to the session. The company's internal stakeholders will be the main topic of the first section. Functional managers, operations staff, finance staff, warehouse and material handlers, and any other employee who will be impacted by the project are examples of these stakeholders. The session's second half will be devoted to discussing external stakeholders. These might be partners, suppliers, test clients, partners' organizations, or anybody living outside of the company.

9.4 Challenges Encountered During Development

It's possible that managers and staff will oppose the implementation of new performance management procedures. Change can be greeted with mistrust, anxiety about the unknown, or worries about how it will affect current processes. Ambiguity in performance expectations and goal-setting can also cause confusion and annoyance. It is challenging for workers to match their efforts with company goals if they are unsure of what is expected of them.

This section provides insights into how our HRM system was developed following an Agile Scrum framework. It outlines the roles and responsibilities of team members, the sprint planning and execution process, and how challenges were identified and addressed throughout the project. The iterative and collaborative nature of Agile Scrum facilitates responsive development, ensuring the project is aligned with the needs of stakeholders.

10. Implementation

10.1 Technical Implementations Details

Our performance management system and employee performance analytics platform feature a strong technological foundation designed to ensure reliability, scalability, and efficiency. At the heart of our backend infrastructure lies PHP, a versatile and widely used server-side scripting language known for its performance and flexibility. Complementing this, our frontend is built using HTML, CSS, and JavaScript, offering users an intuitive and interactive experience. This combination of technologies enables seamless integration between the backend and frontend, facilitating rapid development and customization of features. Leveraging PHP's extensive ecosystem and libraries, we efficiently manage complex business logic and data processing tasks. Meanwhile, HTML, CSS, and JavaScript empower us to create dynamic and visually appealing user interfaces that enhance user engagement and productivity. With a focus on security, our stack incorporates industry best practices to safeguard sensitive employee data and ensure compliance with privacy regulations. Additionally, our performance management system is optimized for performance, enabling real-time analytics and reporting to support data-driven

decision-making. This strong technological foundation underscores our commitment to delivering a high-quality and scalable solution that meets the evolving needs of our clients and users.

10.2 Tools and Technologies Used

Our performance management system and employee performance analytics platform rely on a set of carefully chosen tools and technologies to support its functionality and usability. In the backend, we utilize PHP as our primary scripting language, known for its versatility and robustness in handling complex business logic. Complementing this, the frontend is developed using fundamental web technologies such as HTML, CSS, and JavaScript, ensuring a responsive and interactive user interface. Additionally, we leverage libraries and frameworks like Bootstrap to streamline development and enhance user experience. For version control and collaborative development, Git and GitHub are employed, facilitating seamless code management and team collaboration. Continuous integration and deployment pipelines are established using Jenkins, automating testing and deployment processes to maintain code quality and accelerate time-to-market. This comprehensive set of tools and technologies enables us to build a robust and efficient

performance management solution that meets the needs of our users and stakeholders effectively.

10.3 Code Integration and Inoperability

Our performance management system and employee performance analytics platform are built upon a solid technological framework aimed at seamless code integration and interoperability. PHP serves as our backend scripting language, offering a versatile and well-supported environment for handling complex business logic and data processing tasks. On the frontend, we leverage HTML, CSS, and JavaScript to craft intuitive user interfaces that enhance user interaction and productivity. This combination of technologies facilitates smooth communication between the backend and frontend components, ensuring that our system operates cohesively. Furthermore, our stack prioritizes interoperability, enabling seamless integration with external systems and services. Through standardized communication methods, our platform can easily exchange data with other applications, allowing for enhanced functionality and expanded use cases. This emphasis on smooth communication and interoperability enables us to create a cohesive ecosystem that adapts to the diverse needs of our users and stakeholders, fostering collaboration and innovation.

10.4 Integration Testing and Debugging

In the development of our performance management system and employee performance analytics platform, we prioritize integration testing and debugging to ensure the reliability and functionality of our application. Integration testing involves systematically combining individual components of our system to verify that they work together seamlessly. We employ various testing frameworks and methodologies to assess the interaction between different modules, APIs, and databases. This process helps identify and rectify any inconsistencies or errors that may arise when integrating different parts of the system. Additionally, we dedicate effort to debugging, employing tools and techniques to identify and resolve issues efficiently. Through testing and debugging practices, we strive to deliver a stable and robust application that meets the needs and expectations of our users.

11. Testing And Quality Assurance

11.1 Testing Strategies And Methodologies:

We validate the smooth integration of the performance management system with other HR systems using our testing strategies and methodology. Test the communication and data flow between the learning management system, payroll, and other integrated systems, and the performance management system itself. To guarantee that new updates are completely tested before deployment, incorporate automated testing, continuous integration, and continuous deployment strategies. Implement continuing testing procedures to maintain the quality of the performance management system.

11.2 Test Cases and Test Data:

To verify various system components, test cases are carried out with the use of particular test data. Test cases and test data work together to provide thorough testing coverage, which makes it possible to identify problems with functionality, performance, and data accuracy. Establish a range of goals, such as time-bound, qualitative, and numerical, to confirm the system's adaptability.

11.3 Test Results and Bug Reports:

Test results and bug reports are used to communicate details regarding the state of the system, possible dangers, and areas in need of development. For difficulties to be resolved quickly and to guarantee the overall quality of the analytics and performance management system, teamwork is crucial.

11.4 Quality Assurance Measures:

Our quality assurance measures that all of the system's data, ratings, and performance indicators are consistent. The credibility of performance information may be impacted by anomalies or discrepancies that are found through consistency checks. Keep a record of the procedures, guidelines, and standards applied to analytics and performance management. By ensuring that all parties involved adhere to uniform procedures, standardization lowers the possibility of mistakes and raises overall quality. Give users thorough instruction on how to operate the analytics tools and performance management system. Users are better able to comprehend the system, enter data correctly, and appropriately analyze performance metrics with the support of clear rules and documentation.

12. Results And Evaluation

12.1 Project Outcomes and Deliverables:

ought to be closely in line with the goals and specifications established at the start of the project. Realizing intended outcomes is facilitated by the successful completion of deliverables. The degree to which project outcomes have been met and the caliber of the deliverables generated are frequently used to gauge a project's success. The observable and quantifiable products or outcomes that are created and supplied as part of the project are known as project deliverables. These are frequently particular objects, records, or finished work that support the accomplishment of project objectives.

12.2 Alignment with Project Objectives:

These objectives ought to be in line with the organization's overarching strategic objectives. Make sure the project's objectives are quantifiable, explicit, and apparent. This clarity offers a foundation for assessing the project's effectiveness and aids in comprehending the intended results. Interact with important parties, such as managers, end users, and HR specialists, to make sure their expectations and points of view line up with the project's

objectives. Adopting performance management solutions successfully depends on stakeholder alignment.

12.3 Stakeholder and User Feedback:

Executives, HR specialists, managers, staff members, and outside parties with an interest in organizational performance can all be considered stakeholders in performance management and employee performance analytics. Their input is essential for ensuring that the system is in line with corporate objectives. Users who interact with the performance management system directly, such as managers and employees, offer valuable insights into the usability, efficacy, and ease of use of the tools and procedures. Feedback can be gathered in a variety of ways, such as through surveys, focus groups, interviews, and casual conversations. It might address things like overall satisfaction, report accuracy, user interface design, and system functionality.

12.4 Lessons Learned:

Understanding the lessons learned can help with understanding customer satisfaction and adoption rates. Throughout the whole lifecycle of performance management and analytics projects, knowing which components of the system were well-received by users and where changes can be made helps enhance future implementations, both good and negative. Project plans, implementation techniques, difficulties encountered, and results attained can all be found in this paperwork. It is essential to comprehend the difficulties and roadblocks that arise when implementing and using performance management systems. The identification of problem areas through lessons learnt enables businesses to design solutions for dealing with such issues in the future.

13. Conclusion

In closing, the development of the human resources management system represents a significant milestone in our commitment to improving administrative processes, enhancing educational experiences, and aligning with our client's strategic objectives. The project's adherence to Agile Scrum principles, enterprise architecture best practices, and active stakeholder engagement has resulted in a system that empowers our human resource management community and paves the way for future enhancements. We look forward to the continued evolution and success of our human resources management system, shaping a brighter future for our institution and its community.

13.1 Key Takeaways and Summary

In key takeaways and summary, human resources management is a complex subject that is essential to any organization's success. The significance of strategic planning, economical resource allocation, and successful risk management in achieving financial objectives and guaranteeing long-term sustainability are among the most important lessons learned. Retaining trust and competitiveness also requires embracing technology developments, following rules, and keeping open lines

of communication with stakeholders. Businesses may negotiate obstacles, grab opportunities, and maximize performance in changing economic circumstances by putting strong human resources strategies into place. In the end, in today's intricate business environment, putting a high priority on strong financial management procedures creates the groundwork for stability, profitability, and growth.

13.2 Project Achievements and Contributions

In human resources management Project Achievements and contributions are a good way to measure successful strategic planning and efficient implementation in human resources management. Reaching budgetary goals, making the best use of available resources, and improving operational effectiveness are frequently considered key achievements. Human resources management contributions can take the form of enhancing risk management procedures, creating creative financial models, or putting cost-cutting initiatives into action. Projects significantly contribute to the overall financial health and performance of a business by exhibiting observable benefits like greater profitability, decreased expenses, or enhanced cash flow. Furthermore, by finding new revenue streams, growing market prospects, or

bolstering financial controls, they might potentially support long-term growth. Teams are encouraged when project accomplishments are celebrated and team members are acknowledged for their contributions. This also serves to emphasize the significance of proactive human resources management techniques in fostering organizational success.

13.3 Future Work and Enhancements

Future work in this field will likely focus on leveraging emerging technologies such as artificial intelligence, machine learning, and block chain to streamline processes, improve decision-making, and enhance risk management. Additionally, there will be a greater emphasis on sustainability and ethical investing, driving the adoption of environmental, social, and governance (ESG) principles in financial strategies. Furthermore, advancements in data analytics and predictive model will enable financial managers to anticipate market trends, identify opportunities, and mitigate risks with greater precision. Continuous education and professional development will be essential for staying abreast of industry trends and regulatory changes, ensuring that financial professionals remain equipped to navigate an increasingly complex and interconnected global economy.

13.4 Closing Remarks

In closing, Organizations' success and resilience are built on the foundation of sound human resources management. From forecasting and budgeting to risk management and strategic planning, it covers a wide range of tasks. The role of human resources managers in guiding firms toward their goals while navigating opportunities and difficulties grows more and more important as markets and technology change. Human resources management protects assets and maximizes returns while also building trust among stakeholders and promoting sustainable growth through the promotion of an open, accountable, and innovative culture. As we advance, it is critical that we maintain our flexibility, agility, and forward-thinking while welcoming change and utilizing data and technology to inform strategic decision-making. Human resources management will continue to advance with a steady dedication to quality and honesty.

Our system will continue to develop and growth, resulting in a better and more successful future for its stakeholder and specially in our organization.

14. References

14.1 List Of Sited Sources And References

Here`s the list of references and resources.

References:

1. John Zachman (1987). The Framework for Enterprise Architecture.
2. Schekkerman (2004). Shoenbird incubation behaviour and its influence.
3. Lankhorst et al. (2011). Enterprise Architecture at Work
4. Darrell, et al., (2018). Adversarial Feature Learning
5. Sharman et al., (2016) Measuring Employee Perception of Performance Management System
6. Zhong et al., (2016). Cobalt Carbide Nanoprisms for direct Production
7. Cappelli and Tavis (2018). HR Goes Agile
8. Rigby, et al., (2018). Agile at Scale
9. Barton, et al., (2018). The influence of Social Media usage
10. Audenaert et al., (2019). When employee performance management affects individual innovation in public organization.

These references provide a comprehensive list of sources that have informed the development of the human resource management system project, covering aspects of Agile Scrum methodology, enterprise architecture, relevant studies, and information system integration within higher education.

15. APPENDICES

Supplementary Material:

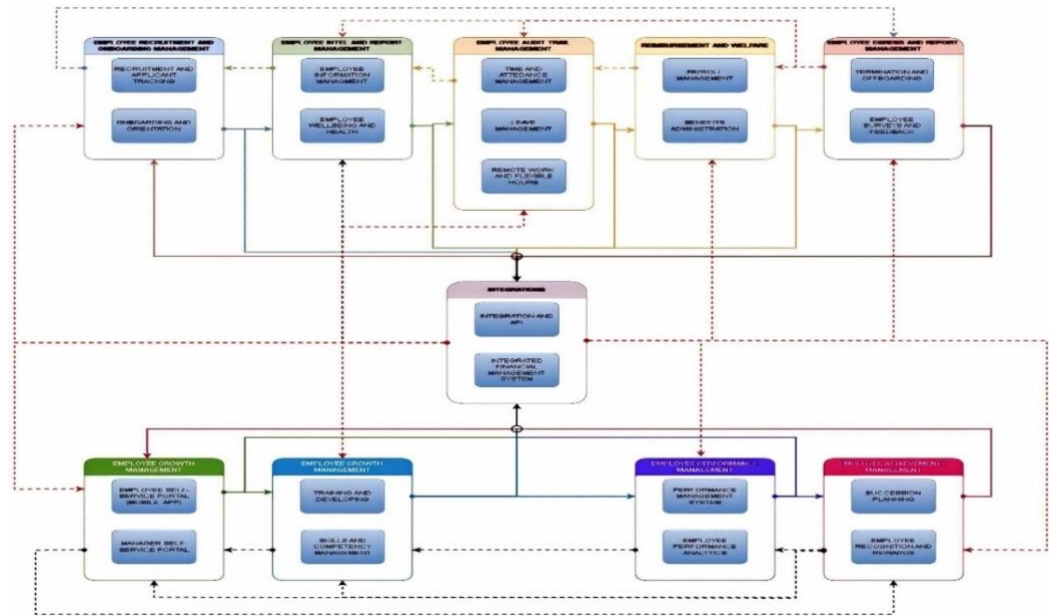


Figure 5. Supplementary Material (BPA2)

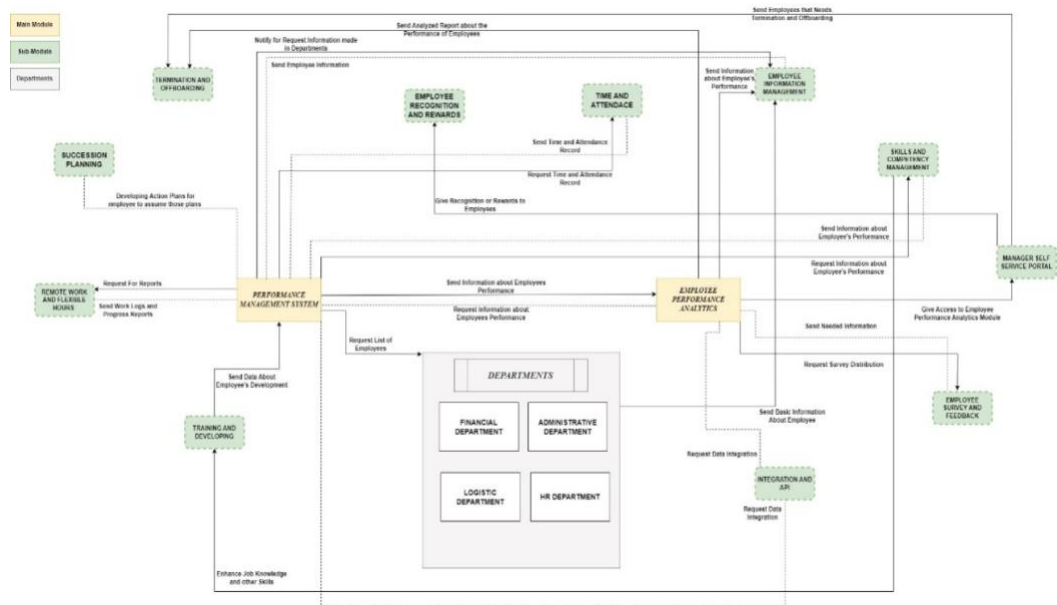


Figure 6. Supplementary Material (BPA1)

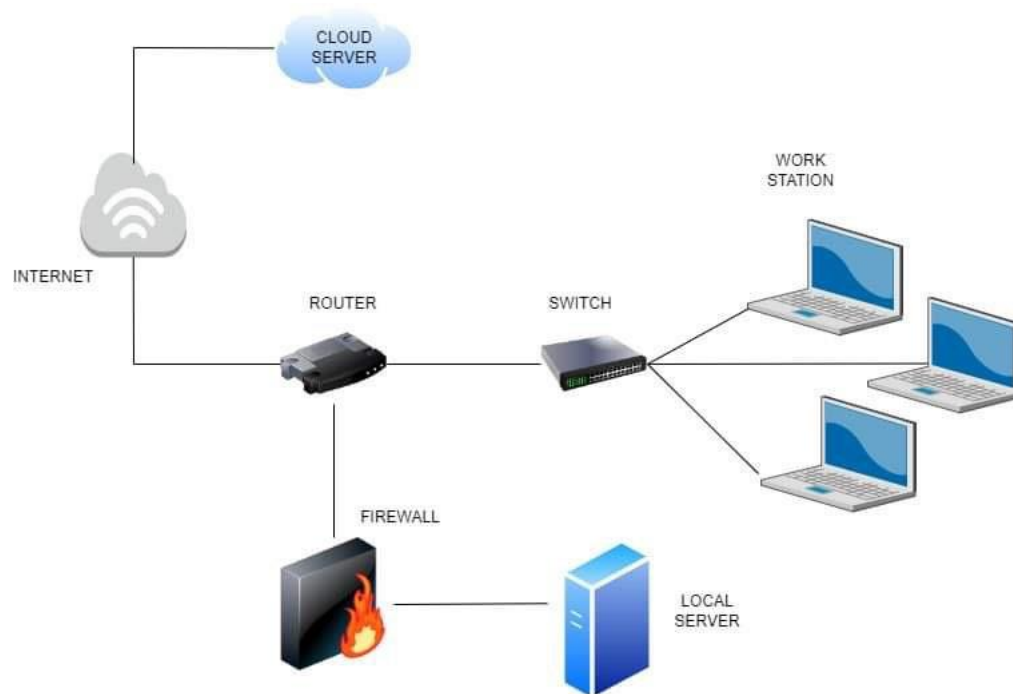


Figure 7. Network Architecture

SPRINT BACKLOG

User Story No.	USER STORIES	User Story Priority	Requirements Reference	Revision Priority	Status
1	As an HR ADMIN , I can log in accountable System.	1	PM-REQ-01	5	DONE
2	As an HR ADMIN , I can analyse the absentees of the employees	1	PM-REQ-02	5	DONE
3	As an HR Admin , I can view activity logs	1	PM-REQ-03	4	DONE

4	As an HR Admin , I can release Employees monthly task reports	1	PM-REQ-04	5	DONE
5	As an HR Admin , I can send them to training if needed	1	PM-REQ-05	4	DONE
7	As an HR ADMIN , I can track employee work hours and overtime	1	PM-REQ-07	3	DONE
9	As an HR ADMIN , I can update personal information.	1	PM-REQ-09	5	DONE
10	As an HR ADMIN , I will provide security and data protection.	1	PM-REQ-10	5	DONE

11	As an HR ADMIN , I can log out.	1	PM-REQ-11	5	DONE
12	As an Employee , I Can log in using OTP through E- Mail	1	EMM-REQ-01	5	DONE
13	As an Employee , I can view the dashboard.	1	EMM-REQ-02	4	DONE
14	As an Employee , I can Request Leave	1	EMM-REQ-03	3	DONE
15	As an Employee , I can Request Leave	1	EMM-REQ-04	3	DONE
16	As an Employee , I can view Pay-slip	1	EMM-REQ-05	5	DONE

17	As an Employee , I can create appointments	1	EMM-REQ-06	5	DONE
18	As an Employee , I can view articles and Programs	1	EMM-REQ-07	5	DONE
19	As an Employee , I can view articles and Programs.	1	EMM-REQ-08	5	DONE

Table 7: Sprint Backlog

Appendix A: Process Diagram

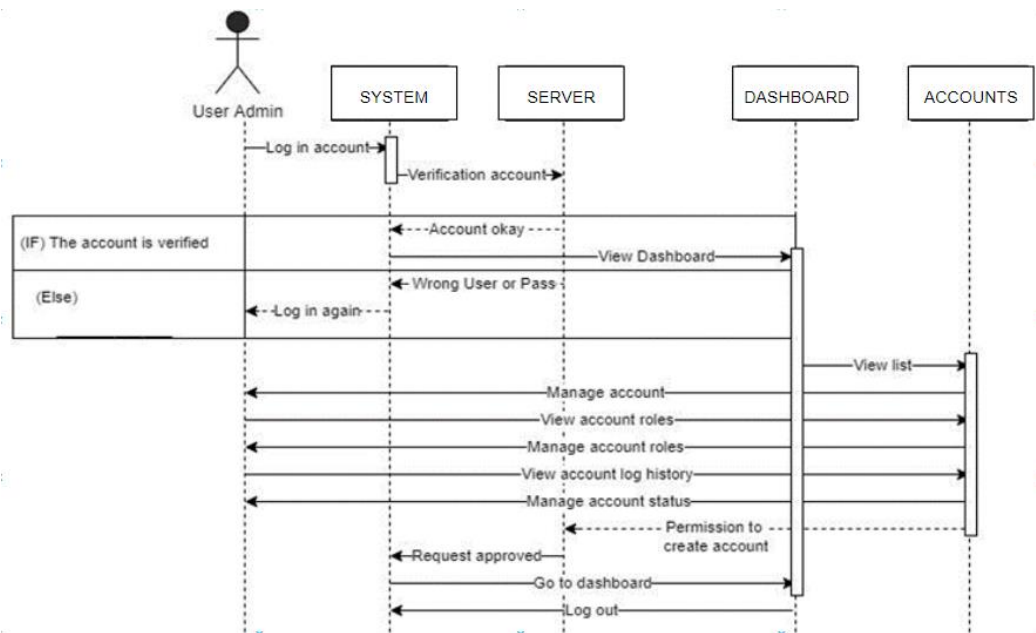


Figure 8: Appendix A: Process Diagram

Appendix B: User Surveys

The following user surveys were conducted to gather feedback on the Human Resources Management System. Participants were asked to provide their opinions and suggestions regarding various aspects of the system to aid in its improvement and optimization.

Survey Questions:

1. How would you rate the ease of use of the Human Resources Management System?
2. How would you rate the overall performance of the Human Resources Management System in terms of speed and responsiveness?
3. Have you experienced any system crashes or downtime while using the Human Resources Management System?
4. Have you encountered any issues with system performance? If so, please describe.
5. Are there any features you feel are lacking or could be improved?

Appendix C: User Stories and Use Cases

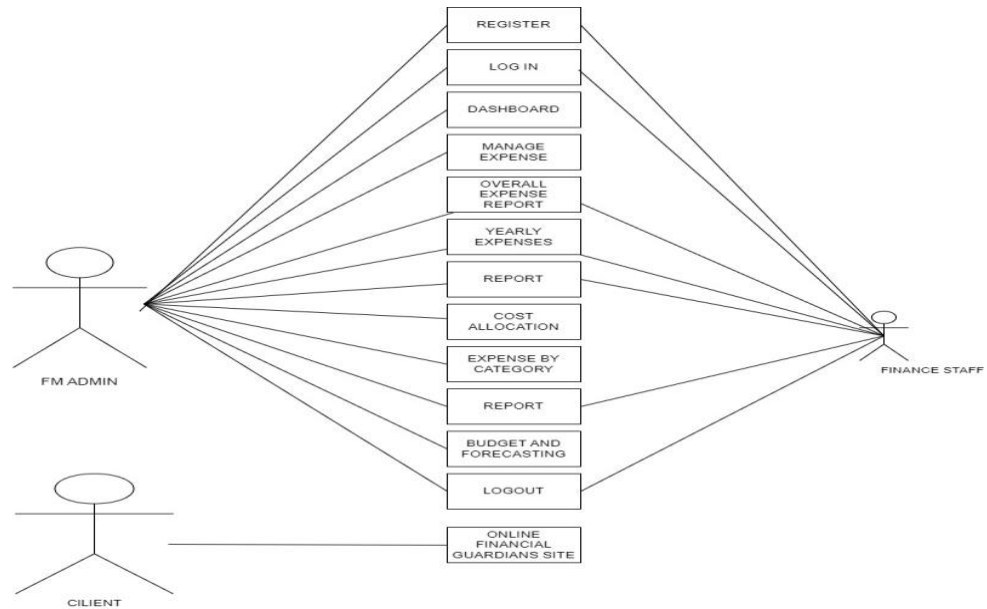


Figure 9. User Stories and Use Cases

Appendix D: Sample Data

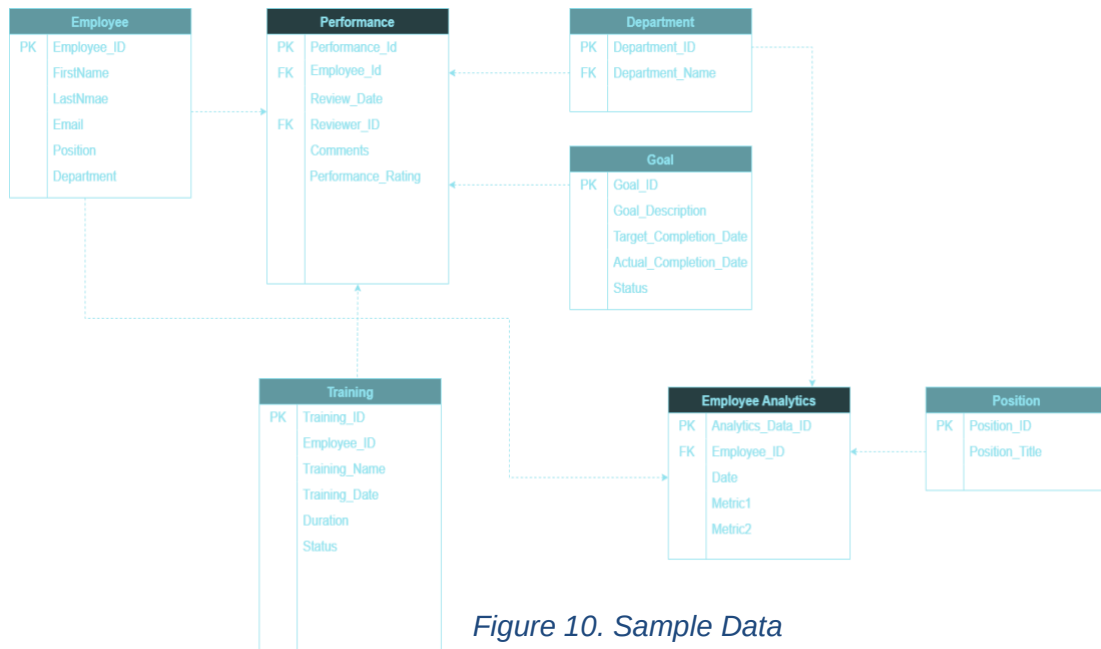


Figure 10. Sample Data

Appendix E: Stakeholder Interview Transcripts

Following Questions:

If have a system:

1. Does your organization have a existing HRMS software that you use to manage and control all HR data?

-NO

If not have system:

1. What tool does your organization used to manage all data and Information of every employee?

- Manually - time in/ time out | using dtr - record details using book

2. Ask to see the hr organizational chart, and their responsibility, their workstation, observed thoroughly.

- declined due to data privacy - provided chart organization

3. How does the organization manage the payroll of every employee?

- manually | collecting the dtr by the end of the month| compile and summarize into excel file | individual check for late/absents (with vl[no salary deduction/ if no vl w/ deduction])/overtime | once finalize

submit to accounting - payroll cut off is every 15th of the month all records should be provided before 15th of the next month | record collection end of the month

4. How does the organization manage the employee off boarding of every employee?

- manually/ due to education set up at the end of the quarter every school year hr provides letter of intent with a question "do you want to stay in our institution for next school year" | letter of intent answering no consider as resignation letter answerable by yes or no
○if no automatically at the end of school year/graduation day consider as last day of report - after of graduation processing of terminated employee submitting grades of the students at registrar return books/ all accountability

5. Does the organization have a employee survey and feedback incase one employee got terminated rom it's position?, if so, how do you manage the survey's and feedback?

- yes/ manually fill up form| part of exit interview questions: best part of being employee list part of being employee self ranking as a employee points of improvement

6. How does the organization manage and analyze the performance and information of every employee?

- assist and analyze by the observation of department head| all employees are under in a certain departments department lower elementary higher elementary junior high school senior high school overall lead by principal year end evaluation teacher evaluation by dept head ➡ forward to principal - to check performance/ top employee - evaluation from students for the performance of teacher.

7. How does the organization manage the employee time and attendance?

- dtr

8. How does the organization manage those employee who are working remotely?

- no attendance via messenger group chat / checking time in and out - hybrid education mtw - school teaching th/f - wfh

9. How does the organization manage the employee recruitment, onboarding and orientation?

- recruitment ○job street (main) ○work of mouth/ referring ○fb page of qc hiring - orientation ○termed call before summer time "march -

may hiring" during/ after pandemic - "june - july" - 2 weeks before class newly hired taking inset (in-service training) oorientation - policies| responsibility | expectations performance incase hired personel doesn't want to continue hr providing different programs for backup employees - not inset anymore/ individual work.

10. Ask to see the reports of every employee contains in the hr organization.

- declined due to data privacy - provided a copy of chart organization

11. When all is well, concluded the interview, and transmit all data to all members.

Appendix F: System Performance Test Results

Simulated typical user workflows including expense entry, budget creation, report generation, and forecasting. System exhibited resilience under stress, with response times increasing gradually beyond 10 concurrent users. No system crashes or failures were observed, although some minor degradation in response time was noted. The system's performance testing indicates that it is well-equipped to handle the financial management needs of our organization. However, ongoing monitoring and optimization efforts are recommended to ensure continued performance under increasing usage and data volumes.

Appendix G: Data Security Measures

The Human Resources Management System will apply in order to secure data and information against unauthorized access, use, disclosure, destruction, alteration, interruption, or distribution, security is the on-going practice of exercising due care and diligence. In this way, we can be aware of information security threats. Human Resources Management System will implement Role-based access control, A set of guidelines used to decide who

gets access to a restricted area or limited information. Then existing firewall, data encryption, two way factor and login authentication. The Security Administrator will be involved in all phases of design, testing, implementation, and migration. Additionally, once the finance system migrates to operational use, it will be monitored by the Security Administrator, along with other IT Tools, to ensure continued compliance with security policies.

The Technical Security Control consists of four (4) branches of policy namely IT Infrastructure Security, Software Security Management, Cloud Security, and Cyber security. These policies prevent vulnerabilities 181 in hardware/software, server and databases, and to the asset of Human Resources Management System.

Appendix H: Mobile Application Prototype

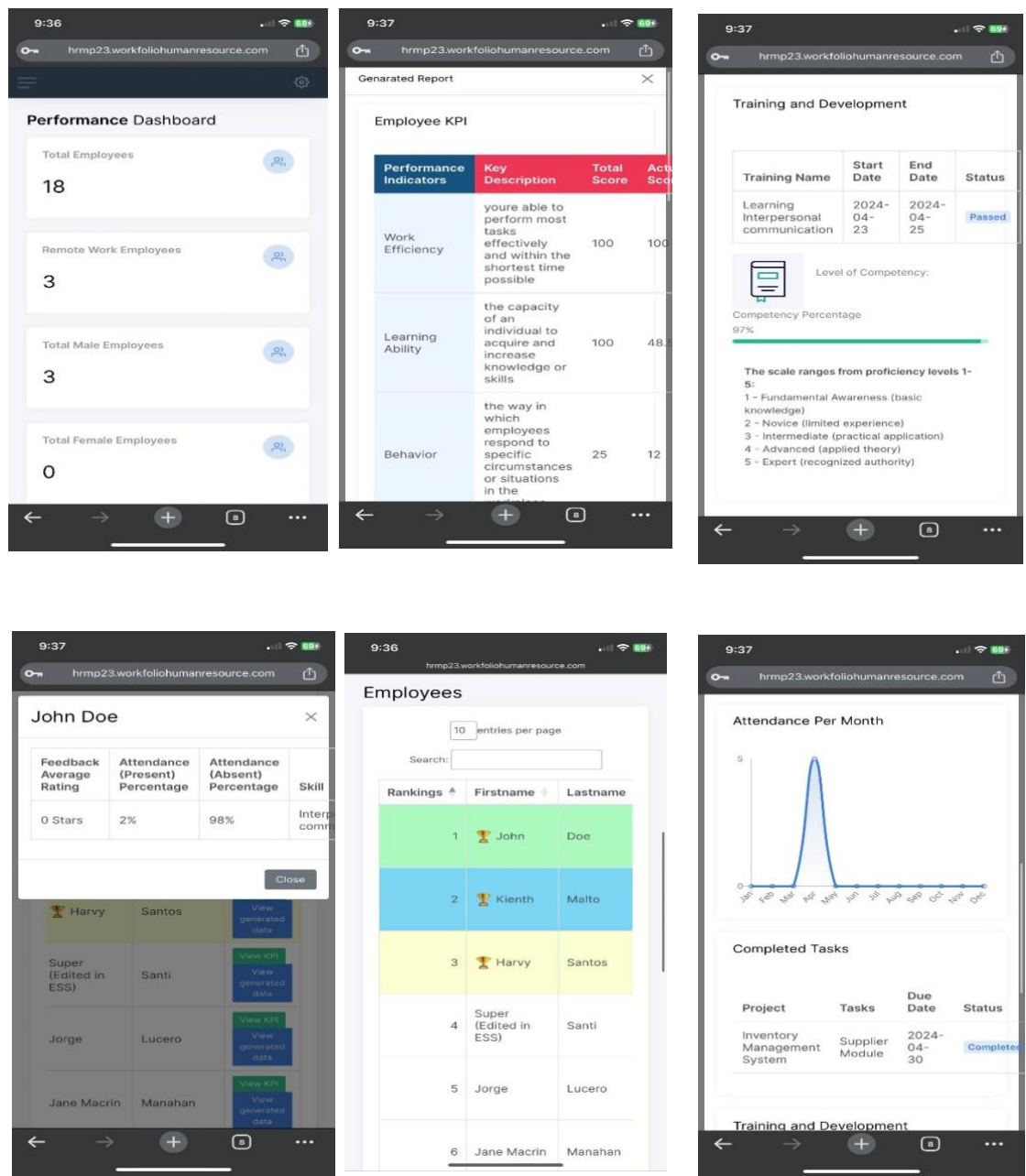


Figure 11. Mobile Application Prototype

Appendix I: Adviser Acceptance (Functional)

The researchers of this group certify that this project study does not incorporate, without acknowledgement, any material previously submitted for a degree of Diploma in any University and to the best of our knowledge and belief. It does not contain any material previously published or written by another person or ourselves except where due reference is made in the text. we also hereby give consent for our Project Study. If accepted, to be made available for photocopying and for interlibrary and for the title and summary to be available to outside organizations.

Signature of Group/ Individual:

Date: March 2024

JOHNLESTER FRANCISCO

JUSTINE ALLEN SEDANO

ANGELA NICOLE L. PALOMO

DENNIS SAGUISA

ARVINJAY PAJULAS

Counter Signed By:

Mr. Jorge B. Lucero

Project Adviser

Appendix J: Panel Evaluation and Signature (plus photo ops during defense)



Figure 12. Panel Evaluation and Signature

Appendix K: Capacity Planning (Estimates on Storage Consumption – 5 years)

The capacity plan for storage consumption over the next five years provides a roadmap for accommodating the data growth of the Human Resources Management System.

Year 1: 13.2 TB

Year 4: 22.814 TB

Year 2: 15.84 TB

Year 5: 27.379 TB

Year 3: 19.008 TB

Recommendations:

- Provision additional storage capacity in alignment with the estimated growth rates to ensure uninterrupted system operations.
- Regularly monitor actual data growth and adjust capacity planning strategies accordingly.
- Consider implementing data management techniques such as compression and deduplication to optimize storage utilization.
- Evaluate the feasibility of leveraging cloud storage options for scalability and cost-effectiveness beyond the initial planning period.
- Continuously review and update the capacity plan to adapt to evolving business requirements and technological advancements.

Appendix L: Pilot Companies' Background with proofs of interviews



Figure 13. Pilot Companies' Background with proofs of interviews

Appendix M: Cloud Copy of the Codes (1 year validity)

<https://github.com/Sheshable/Lester.git>

Appendix N: IMRAD Format Summary

INTRODUCTION

Human Resource Management is a strategic orchestration of an organization's most valuable asset: its people. It is that element that gives an organization the most crucial tasks of bringing in, creating, developing, retaining, and leading its employees to institutionalized long-term objectives but in this research, we tackle about the Performance Management and Employee Performance Analytics

Based on the data, the most associated problem occurs in the performance of the employee is Insufficient Training —a lack of training leads to a number of workplace issues. The most obvious one is that employees aren't able to do their jobs as well as they would like, but it goes deeper. Second lack of communication, this is a common problem in the workplace. But without communication, it's difficult for employees to know what's expected of them. It also means that other problems go unreported, rather than getting fixed, they begin to stack up. Lastly, little recognition It's natural that people want to be recognized for the hard work they do. It's easy to get into the mindset that an employee's pay is their recognition, but often it's the small details that really matter. When someone does great work, it's an excellent opportunity to boost morale and build trust, so don't miss out on it.

METHOD

The process we have used in developing the system is Agile Methodology: Scrum. We first analyzed the data we have gathered and understand the process on how the data of an employee is processed inside an organization when the employee started doing their responsibility within the organization. We also analyzed the employee's performance based to their attendance and the workloads that they've accomplished. After analyzing those data, We start allocating and

constructing Product Backlog Items and inserted them in our Sprint, alongside our Daily Standup meeting and Sprint Retrospectives.

RESULT

The data we gathered gives us a advantage of how we will construct our system, and give us all the data we need to design the system architecture and processes that we need to achieve in order to deploy the system online. Furthermore, the data we gathered give us also a knowledge of how to construct possible features that we may implement in our system

DISCUSSION

The integration of performance analytics allows organizations to identify performance trends and patterns, both at the individual and team levels. This data-driven approach enables proactive interventions to address performance gaps and capitalize on strengths. By fostering a culture of continuous improvement, the system facilitates adaptive responses to changing business environments and ensures ongoing organizational resilience.

16. Appendices: Technical Documentation

16.1. System Architecture:

Overview of the overall system architecture

In the Overview of the overall system architecture diagram we explain how our components and modules interact with other modules. How our Components Process a data, how our module create data. We also put the authentication of the system. System Flow, Forms and their process and the Relationship between the Enterprise Integration System and our modules.

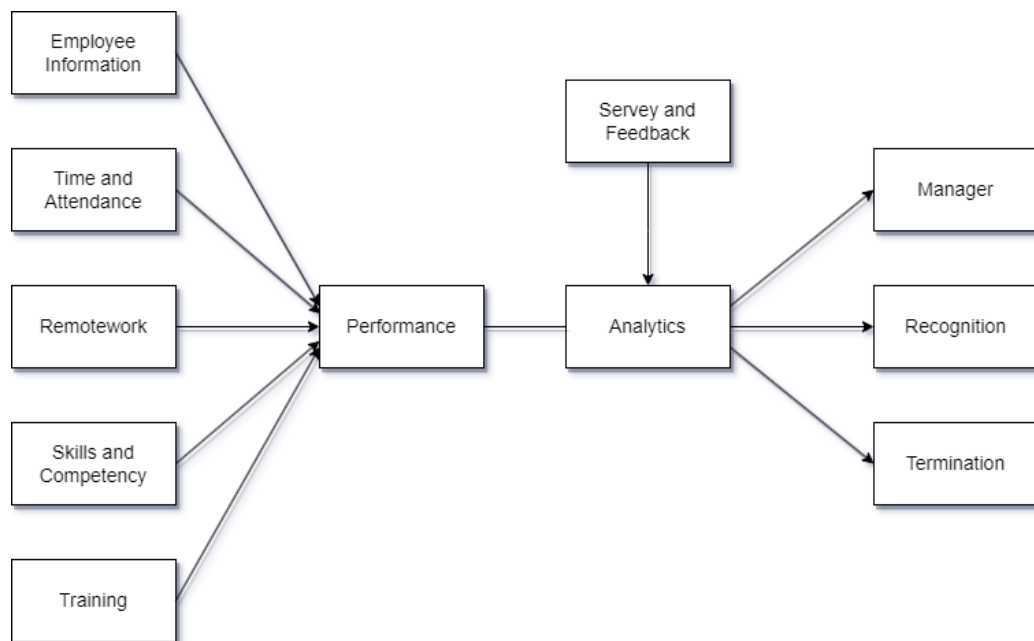


Figure 14. System's Flow

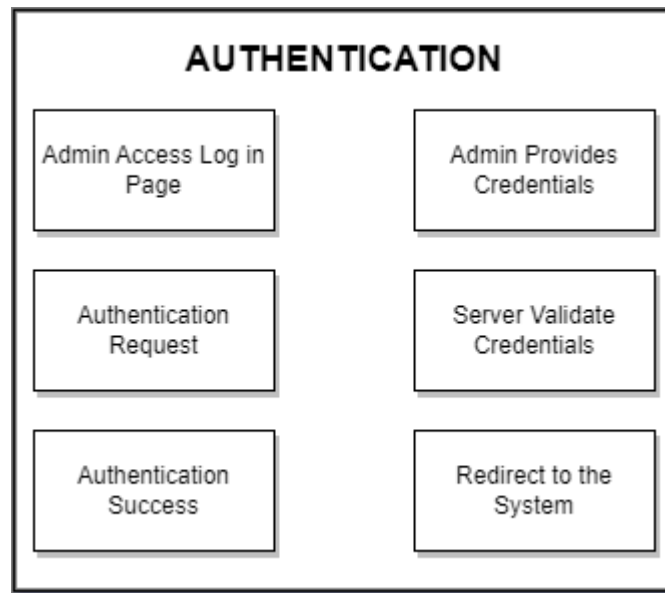


Figure15. Authentication of the System

Rankings	Firstname	Lastname	Tools
1	John	Doe	View KPI View generated data
2	Kienth	Malto	View KPI View generated data
3	Harvy	Santos	View KPI View generated data
4	Super (Edited in ESS)	Santi	View KPI View generated data
5	Jorge	Lucero	View KPI View generated data

Figure 16. KPI

16.2 INFORMATION SYSTEMS INTEGRATION

Description Of How The Class's Information Systems Were Integrated

One of the main benefits of HRIS is its capacity to centralize large amounts of data about employees. Paper-based methods of keeping records or using a variety of software systems frequently result in errors and inefficiencies. HR Information System is a central platform that allows all pertinent employee data, such as review of performance as well as training history as well as career plans, to be kept in a safe and easy-to-access manner. The central repository not only reduces time but also assures precision and uniformity in the information, creating an excellent foundation for effective development and performance management.

DIAGRAMS SHOWING THE INTEGRATION POINTS AND DATA FLOWS

- Web-Based Application: User interface for HR professionals, recruiters, and new hires.
- Integration Layer/APIs: Facilitates communication and data exchange between system components.
- HRIS: Central repository for employee data.

Data Transformation And Mapping Processes

PHP (Hypertext Preprocessor): PHP can be used for backend data transformation processes. For instance, you can write PHP scripts to transform data fetched from databases or external APIs into a format suitable for presentation on web pages. PHP can also facilitate data mapping by processing incoming data, validating it, and mapping it to appropriate data structures within the application's logic.

Native Development (e.g., Java for Android, Swift for iOS): In native mobile development, data transformation occurs within the application logic. Native code processes and transforms data fetched from servers or local databases to populate user interfaces. Mapping processes in native development involve mapping server responses or local database records to corresponding UI components, such as lists, tables, or forms.

CSS (Cascading Style Sheets): While CSS primarily handles the presentation and styling of web content, it indirectly influences data transformation by defining how data is displayed to users. CSS can be used to format and style data tables, lists, and other HTML elements to enhance readability and user experience during data presentation.

JavaScript (JS): JavaScript plays a crucial role in client-side data transformation and manipulation within web applications. It enables dynamic updates to web page content based on user interactions or data changes. JavaScript libraries and frameworks like jQuery, React, or Vue.js facilitate data binding, filtering, sorting, and formatting operations, contributing to seamless data transformation and mapping on the client side.

Bootstrap: Bootstrap, as a frontend framework, provides prebuilt components and styles for creating responsive and visually appealing web interfaces. While Bootstrap itself doesn't directly handle data transformation or mapping, it enhances user interface design, making data presentation more effective and intuitive.

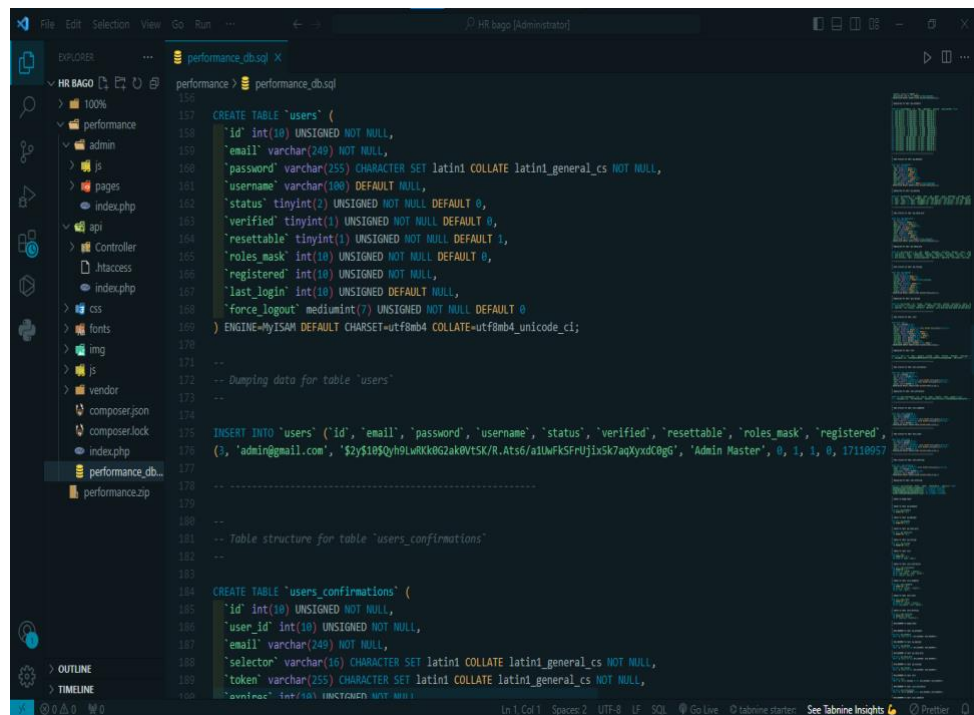
16.3 APPLICATION DESIGN AND DEVELOPMENT

Detailed Information About The Software Modules And Components

The Human Resource Management System (HRMS) is composed up of various software components and modules, each of which performs a certain job inside the system. The foundation of the HRMS is the employee information module, which is in charge of organizing and maintaining a wide range of employee data, such

as personal profiles, job descriptions, and performance reviews. The automatic time and attendance reporting system is integrated with this module. It tracks employee work hours and produces real time information to help with payroll processing and labor law compliance. Furthermore, managers can design optimal work schedules based on skill sets, business requirements, and staff availability with the help of the shift scheduling process module. This module guarantees effective resource allocation, minimizes conflicts, and expedites the scheduling process.

CODE SNIPPETS AND CODE STRUCTURE



```
154 performance > performance_db.sql
155
156 CREATE TABLE `users` (
157   `id` int(10) UNSIGNED NOT NULL,
158   `email` varchar(240) NOT NULL,
159   `password` varchar(255) CHARACTER SET latin1 COLLATE latin1_general_cs NOT NULL,
160   `username` varchar(100) DEFAULT NULL,
161   `status` tinyint(2) UNSIGNED NOT NULL DEFAULT 0,
162   `verified` tinyint(1) UNSIGNED NOT NULL DEFAULT 0,
163   `resettable` tinyint(1) UNSIGNED NOT NULL DEFAULT 1,
164   `roles_mask` int(10) UNSIGNED NOT NULL DEFAULT 0,
165   `registered` int(10) UNSIGNED NOT NULL,
166   `last_login` int(10) UNSIGNED DEFAULT NULL,
167   `force_logout` mediumint(7) UNSIGNED NOT NULL DEFAULT 0
168 ) ENGINE=MyISAM DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
169
170 -- Dumping data for table `users`
171 --
172
173 INSERT INTO `users` (`id`, `email`, `password`, `username`, `status`, `verified`, `resettable`, `roles_mask`, `registered`,
174 0, 'admin@gmail.com', '$2y$10$Qy9h9LwRk8G2ak0Vt5K/R.Ats6/a1MfK5Fuj1x5K7aqXydc0gg', 'Admin Master', 0, 1, 1, 0, 17118957
175
176
177
178 -- Table structure for table `users_confirmations`
179 --
180
181 CREATE TABLE `users_confirmations` (
182   `id` int(10) UNSIGNED NOT NULL,
183   `user_id` int(10) UNSIGNED NOT NULL,
184   `email` varchar(240) NOT NULL,
185   `selector` varchar(16) CHARACTER SET latin1 COLLATE latin1_general_cs NOT NULL,
186   `token` varchar(255) CHARACTER SET latin1 COLLATE latin1_general_cs NOT NULL,
187   `expire` int(10) UNSIGNED NOT NULL
188 ) ENGINE=MyISAM DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

Figure 17. Sample Code

Algorithms And Data Structures Used

Algorithms and data structures are essential to the Human Resource Management System (HRMS) because they maximize the effectiveness and performance of many functions, such as shift scheduling, automated time and attendance reporting, and employee information management. Data structures like hash tables or linked lists may be used in the employee information module to effectively store and retrieve employee data. Fast and precise data manipulation is ensured by algorithms for personnel record searching, sorting, and updating. Algorithms are used in the automated time and attendance reporting system to determine work hours based on clock in and clock out times. To maximize efficiency, these algorithms frequently make use of interval trees or time-based computations. Data structures like priority queues or arrays can be used to store timekeeping information and generate automated reports.

Libraries And Frameworks Utilized

An array of libraries and frameworks are used throughout the construction of the Human Resource Management System (HRMS) to facilitate the implementation of important features like shift scheduling procedures, automated time and attendance reporting,

and employee information management. Frameworks like Spring or Django offer strong backend support for data storage and management, while libraries like React or Angular can be used to create user-friendly interfaces for data entry and retrieval in the employee information module.

16.4 DATABASE SCHEMA AND DATA MANAGEMENT

Database schema design

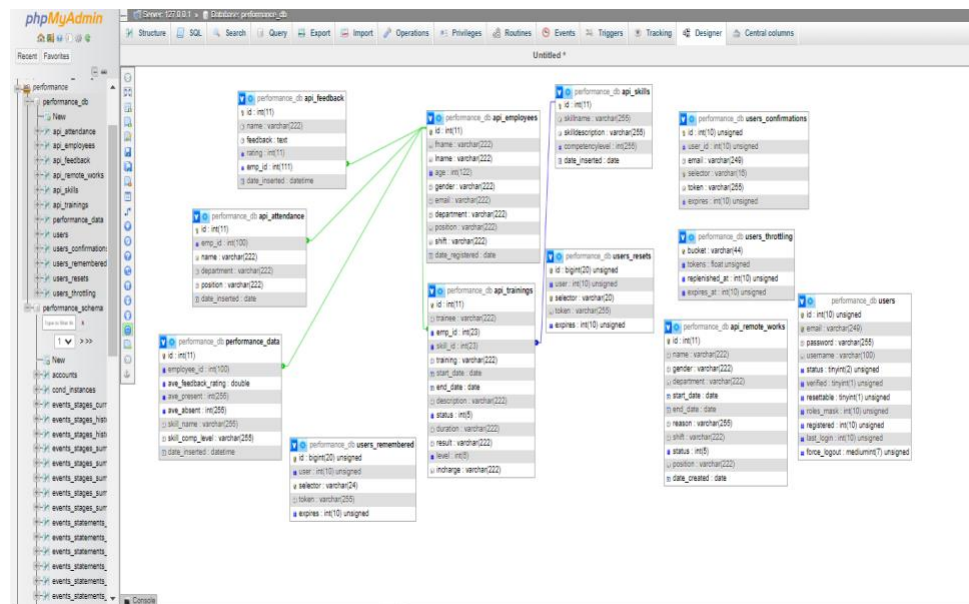


Figure 18. Database Schema

Entity relationship diagrams (ERDs)

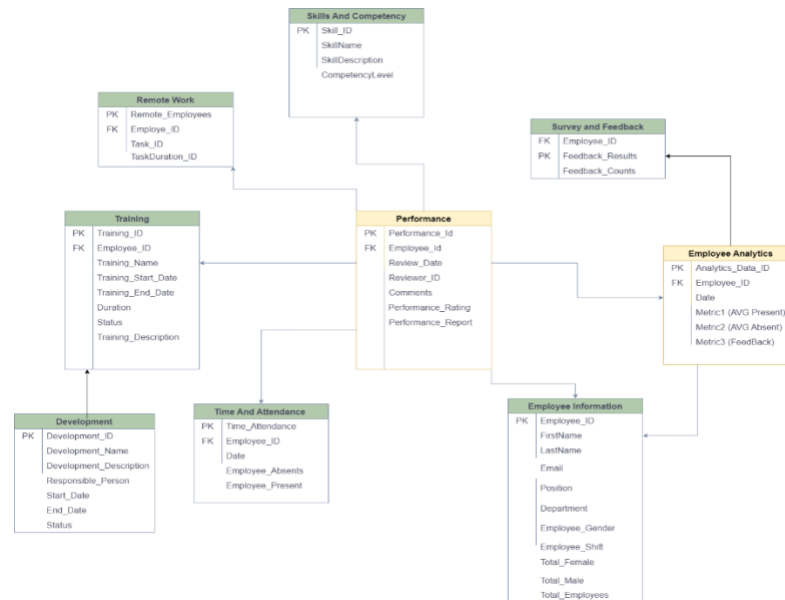


Figure 19. Entity Relationship Diagram

Data Modeling And Normalization

Data modeling Methods such normalization is used to guarantee data accuracy and enhance database efficiency. Normalization improves performance and decreases redundant data by arranging data into logical structure

CRUD (Create, Read, Update, Delete) operations

- **Create:** Insert new records into the table.
- **Read:** Retrieve existing records from the table.
- **Update:** Modify existing records in the table.
- **Delete:** Remove records from the table.

CREATE

```
1 INSERT INTO `performance_data`(`id`, `employee_id`, `ave_feedback_rating`, `ave_present`, `ave_absent`,  
  `skill_name`, `skill_comp_level`, `date_inserted`) VALUES ('[value-1]', '[value-2]', '[value-3]', '[value-  
4]', '[value-5]', '[value-6]', '[value-7]', '[value-8]')
```

READ

```
1 SELECT `id`, `employee_id`, `ave_feedback_rating`, `ave_present`, `ave_absent`, `skill_name`,  
  `skill_comp_level`, `date_inserted` FROM `performance_data` WHERE 1
```

UPDATE

```
1 UPDATE `performance_data` SET `id`='[value-1]', `employee_id`='[value-2]', `ave_feedback_rating`='[value-  
3]', `ave_present`='[value-4]', `ave_absent`='[value-5]', `skill_name`='[value-6]', `skill_comp_level`='[value-  
7]', `date_inserted`='[value-8]' WHERE 1
```

DELETE

```
1 DELETE FROM `performance_data` WHERE 0
```

16.5 Network Configuration

NETWORK TOPOLOGY AND CONFIGURATION DETAILS

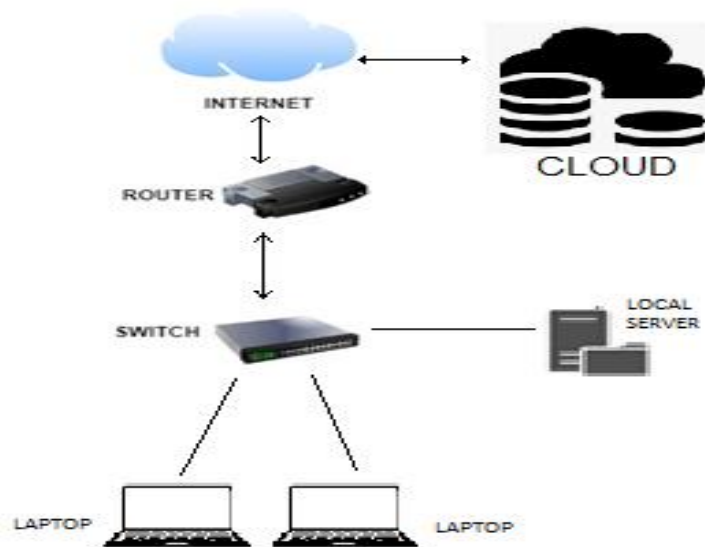


Figure 20. Network Topology

Security Measures (E.G., Firewalls, Encryption)

Human Resources, managing tasks like recruitment, applicant tracking, onboarding, and orientation is crucial for a company's success. Think of it like how Windows Defender protects your computer from viruses and threats. An HR system does something similar but for hiring and welcoming new employees. It keeps everything organized in real-time, making sure job applicants are handled properly, and new hires get the help they need to start smoothly. Just like how Windows Defender updates itself to stay effective, the HR system adapts to changes in rules and best

practices. It works seamlessly with other HR tools, making life easier for HR teams. And the best part? It doesn't overwhelm anyone with extra work. Instead, it helps HR focus on important tasks, like building a great team and making sure everyone feels welcome and supported. So, just as Windows Defender keeps your computer safe, the HR system helps keep the company running Human Resources, managing tasks like recruitment, applicant tracking, onboarding, and orientation is crucial for a company's success. Think of it like how Windows Defender protects your computer from viruses and threats. An HR system does something similar but for hiring and welcoming new employees. It keeps everything organized in real-time, making sure job applicants are handled properly, and new hires get the help they need to start smoothly. Just like how Windows Defender updates itself to stay effective, the HR system adapts to changes in rules and best practices. It works seamlessly with other HR tools, making life easier for HR teams. And the best part? It doesn't overwhelm anyone with extra work. Instead, it helps HR focus on important tasks, like building a great team and making sure everyone feels welcome and supported. So, just as Windows Defender keeps your computer safe, the HR system helps keep the company running

PROTOCOLS AND COMMUNICATION METHODS USED

Protocol /		Description
Communication		
Method		
Email Communication		Fundamental tool for formal communication, document sharing, scheduling interviews, and updates.
Instant Messaging Platforms		Realtime communication for quick questions, updates, and coordination among HR team members.
Video Conferencing		Facilitates virtual meetings, interviews, and group discussions, especially for remote hiring.
Document Sharing and Collaboration Tools		Cloud based platforms for storing, sharing, and collaborating on recruitment and onboarding documents.
HR Management Systems (HRMS)		Centralizes HR related communications, documentation, and workflows in one integrated platform.

Project Management Tracks recruitment projects, coordinates Tools tasks, and manages deadlines for HR team members.

Regular Meetings and Scheduled team meetings and one-on-one Check-ins check-ins to discuss progress, address challenges, and brainstorm solutions.

Feedback and Structured sessions for reflecting on Evaluation Sessions processes, identifying areas for improvement, and implementing changes.

Load balancing and failover mechanisms

Redundant Systems, Load balancer.

16.6. Deployment and Infrastructure:

Deployment strategies (e.g., cloud, on-premises)

When making the network and the system, you must employ various kinds of hardware. This hardware is referring to the external and internal devices all equipment enable you to perform major functions such as input, output, storage, communication, processing, and more. It helps to establish an effective mode of communication, thereby improving the facilitating of the business process.

Hardware specifications and server configurations

User tools Computers, laptops, routers, server, and gateways are a few examples of the devices that make up a network. Every network architecture's objective, in a sense, is to determine the most effective method of transferring data from one hardware point to another.

Configuration management (e.g., scripts, automation tools)

In our system, wires or cables are utilized to transfer data signals through a wired transmission medium. This type of communication is referred to as "wired."

Unshielded Twisted Pair (UTP) Shielded Twisted Pair (STP). The goal of STP is to eliminate electromagnetic interference by conducting a shield across the four pairs of signal-carrying wire. The cable that is used to connect devices in a local area network. It has four pairs of cables that are each twisted with a variable number of pairs per inch to reduce interference. This configuration is frequently referred to as UTP.

Scalability and resource optimization

- Server – 127.0.0.1 via TCP/IP
- Server type – MariaDB

- Services – Apache/ 2.4.54 (Win64) OpenSSL/1.1.1p PHP/8.2.0 mySQL 8.2.0.
- Network Configuration – IP address, Subnet mask, Gateway, and DNS server
- Security - Firewall

16.7. Security Measures:

Security protocols and measures implemented

When it comes to infrastructure security, it typically means measures to protect infrastructure, such as network communications, communication center, server center, database center and IT center. Infrastructure security seeks to limit vulnerability of these structures and systems from sabotage, viruses and contamination.

These are the examples of infrastructure security implemented within the system:

- Secure Socket Layer (SSL) – standard technology for keeping an internet connection secure and safeguarding any sensitive data that is being sent between two systems, preventing criminals from reading and modifying any information transferred including potential personal details.
- Cloud Server Management – refers to exercise of control over public, private or hybrid cloud infrastructure resources and services. Cloud

server has been in-demand for over the years because of its reliability, security and convenience.

Authentication and authorization mechanisms

Login Authentication

- The most popular user authentication mode requires users to submit a unique username and password to access the system.
- A server uses authentication to determine who is accessing their website or data.
- Two-factor authentication (2FA), which provides an extra layer of security, includes utilizing two distinct forms of identity: a password and a security token.
- Authentication does not limit a user's ability to access a particular file or perform a precise operation. Authentication is the only method to identify and confirm a system or a person.
- Encryption and data protection

Access control is the most common way of conceding or denying clients admittance to explicit assets or activities inside a business framework. This is vital to data security since it guarantees that secret information is defended and that prominent approved people get to significant business processes.

Incident Response And Mitigation Procedures

In case such an incident that would happen directly or indirectly to the system, an incident response team will be deployed in order to contain any damage to the system.

- Incident commander

Responsible for driving the incident to resolution and facilitating activities during incident. He/she has overall responsibility for managing the incident by establishing objectives, planning strategies and implementing recovery tactics.

- Subject Matter Expert

The developers of the system are being investigated. We provide the knowledge and expertise in the technical area for the project.

- Customer Liaison

Interacts with the customers to keep them accurately informed about the status of the incident that happened. An effective liaison is approachable and persuasive.

- Scribe

Record important information and decisions upon the incident. This frees the project manager to focus on leading the discussion, resolving conflicts or keeping the participants on track. This might involve recording the meeting, taking notes online or making notes on a paper and transcribing them later.

16.8. Testing and Quality Assurance:

Unit Testing

Can confirm the accuracy of algorithms used to compute performance indicators, like sales quotas fulfilled or project completion rates. This guarantees the reliability of the performance information given to managers.

Integration Testing

To guarantee smooth data interchange and compatibility, integrations testing with other systems or data sources, such as HR databases or performance review platforms.

Performance Testing

Testing the system's capacity to grow or shrink in response to demand fluctuations. This entails evaluating the effects of adding servers or database capacity on performance in order to make sure the system can grow in the future without experiencing performance loss.

Compatibility Testing

Testing the program on various web browsers will guarantee that functionality and performance are consistent across all platforms. This

guarantees that managers and employees won't run into compatibility problems when using their favorite browsers to access the system.

Feedback Loop

Make sure that the systems for recording and capturing feedback from peers or supervisors are working properly by testing them. This involves confirming that the appropriate employee and performance indicator is linked to the feedback.

16.9. System Monitoring and Maintenance:

Performance Monitoring

Organizations can continuously enhance employee performance by iteratively refining their strategy and using performance monitoring data to find areas for improvement in performance management procedures, tools, and actions.

User Activity Monitoring

Keeping track of user logins and session lengths provide information on how often workers access the performance management system and how much time they spend using it each time.

Document Monitoring

Assessing the frequency and caliber of feedback given is made easier by keeping an eye on the feedback document exchanges across peers, supervisors, and staff. This involves keeping track of the completion of performance improvement programs, written comments, and feedback forms.

Security Monitoring

Utilizing encryption methods to protect performance management system data while it's in use and while it's in transit. This guarantees that confidential information about employee performance is shielded from unauthorized access or interception.

Backup Monitoring

Putting in place automated backup processes to make periodic copies of performance management files, databases, and configurations. This guarantees that current data is regularly backed up and accessible for restoration in the event of system failure or data loss.

16.10. APIs and Integration Points

Documentation Of Apis Used For Integration

The documentation of APIs used for integration between various systems such as performance management, analytics, employee information, time and attendance, remote work, training, and feedback. The API documentation serves several important purposes in integrating various systems. It provides clarity on the integration points, outlining the APIs used for communication, data exchanged, and flow. It assists in onboarding new developers, helping them understand the integration quickly. Troubleshooting becomes easier with documentation, providing insights into expected behavior and error resolution. Updates and changes to APIs are documented to keep developers informed. Security guidelines and best practices are included to ensure secure interactions with the APIs. Documentation helps developers optimize performance and comply with privacy and regulatory standards.

Api Endpoints, Methods, And Data Formats

API Endpoints: API endpoints are URLs through which clients interact with an API. They represent distinct resources or functionalities offered by the API. For instance, in a social media API, endpoints could

include /employee for accessing employee data, and /posts for retrieving posts.

Methods: API methods are used to determine the action requested at specific endpoints. Common methods include (GET), Retrieves information. It's like asking for data. (POST), Adds new information. It's like submitting a form. (PUT), Updates existing information. It's like editing an entry. (DELETE). Removes information. It's like deleting a file or record. (Endpoint) is a platform that enables users to perform specific actions (Methods) depending on the contents it interacts with.

Data Formats: APIs typically exchange data in specific formats, such as JSON (JavaScript Object Notation) or XML (eXtensible Markup Language). JSON is widely used due to its simplicity and readability. For instance, a GET request to /users might return data in JSON format

How External Systems Interact With Your Project

APIs (Application Programming Interfaces) are crucial for enabling seamless interaction between external systems and a project. For instance, we have developed APIs to enable third-party applications to access specific functionalities or retrieve data from our system. These APIs follow standardized protocols and formats, allowing external systems to easily understand and interact with them.

Webhooks play a crucial role in web-based projects by enabling smooth communication between various systems. In a project management tool, custom applications can be developed to monitor project milestones and deadlines. Using webhooks, external systems like notification services or analytics platforms can subscribe to key events within the project management tool, such as task completions or deadline updates.

Direct integration scenario, external systems communicate with your project using Application Programming Interfaces (APIs) and integration points tailored for smooth interaction. For example, in an e-commerce platform, payment gateways or shipping carriers can directly engage with your system via APIs for transactions and logistics. APIs grant access to vital functions and data, like secure payment processing or shipping details. Integration points within your project act as bridges, facilitating efficient and dependable information exchange with external systems.

16.11 User manual and guides

16.11.1.1 Instruction for System Users

This manual will guide the users through the functionalities and usage of the system, including logging and out, Performance Management, and Employee Performance Analytics System.

Begin with Log in

- On the homepage, enter your **email** and **password**.
- Click **Submit** to log in

The screenshot shows a web browser window with a login page. The browser's address bar displays a URL: `hrmp23.workfoliohumanresource.com/?fbclid=IwAR26sxyT_xtBoZ0PjAgJjVGo2ugu_CkeuA-6pmH15dnoErVrOyxccuINVAc_aem_AQMucroneXXkxtrj8qp1q_MrHu...`. The browser's bookmark bar includes links to YouTube, Vite + React + TS, Bestlink College of L..., Virtual Learning Lab..., Anime watch, Kit | Font Awesome, (56) Performance M..., Facebook, and Hlgyu/Bago uli at m... The login page features a circular logo at the top, followed by the text "Login to your account" and "Enter your credentials below". Below this, there are two input fields: "Username" with the value "john@doe.com" and "Password" with masked characters "*****". A blue "Sign in" button is positioned at the bottom of the form. To the left of the form, a diagram illustrates the login process with two blue boxes: "Step 1 Input your Email" and "Step 2 Input your Password", each with a blue arrow pointing to the corresponding input field.

- Click **“Profile”** and Click **“Log Out”**

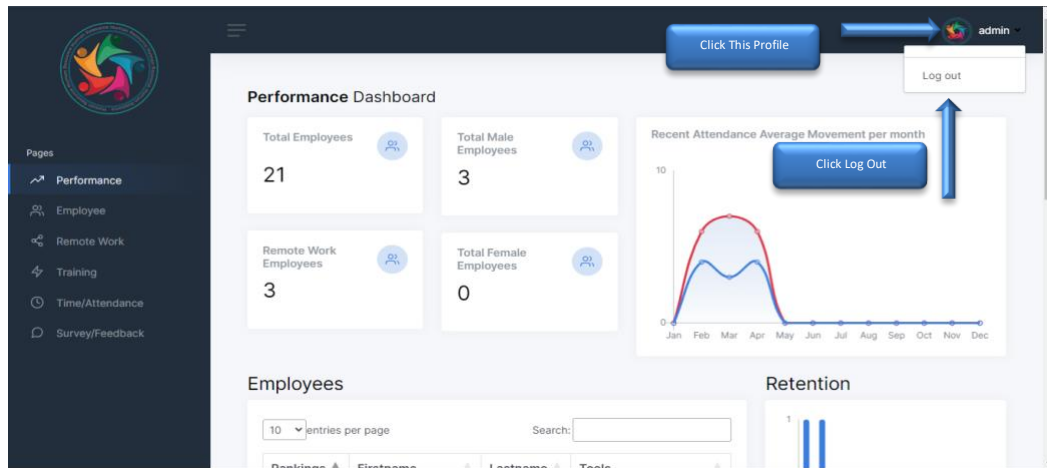


Figure 21: Log In and Log Out Guide

Dashboard for Performance and Analytics

- After you log in, you will see the Performance Dashboard this Dashboard
- **Scroll Down** the screen to see the Employee's Rankings

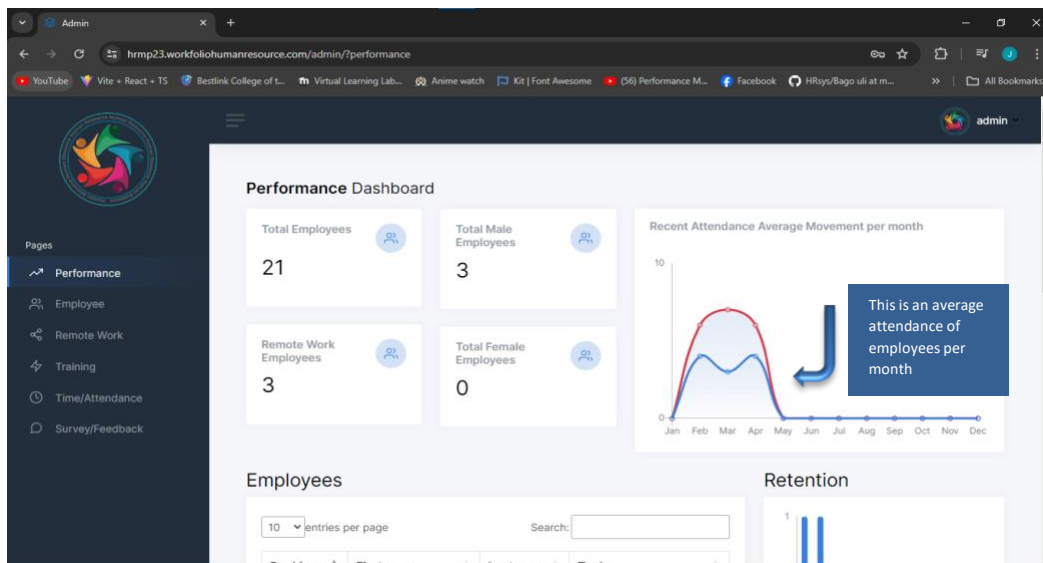


Figure 24: Performance Dashboard

Employees Rankings

- Click the **View KPI** of the employee to see his generated report

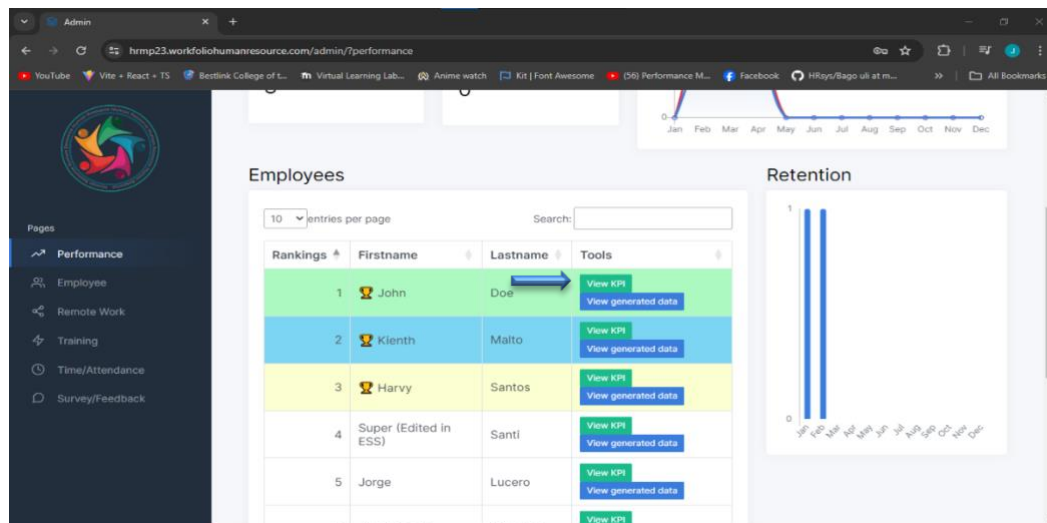


Figure 24: Performance Dashboard 2

- After clicking the **View KPI** you will the **Employee KPI**
- **Scroll down** the screen to see his **Attendance Per Month**

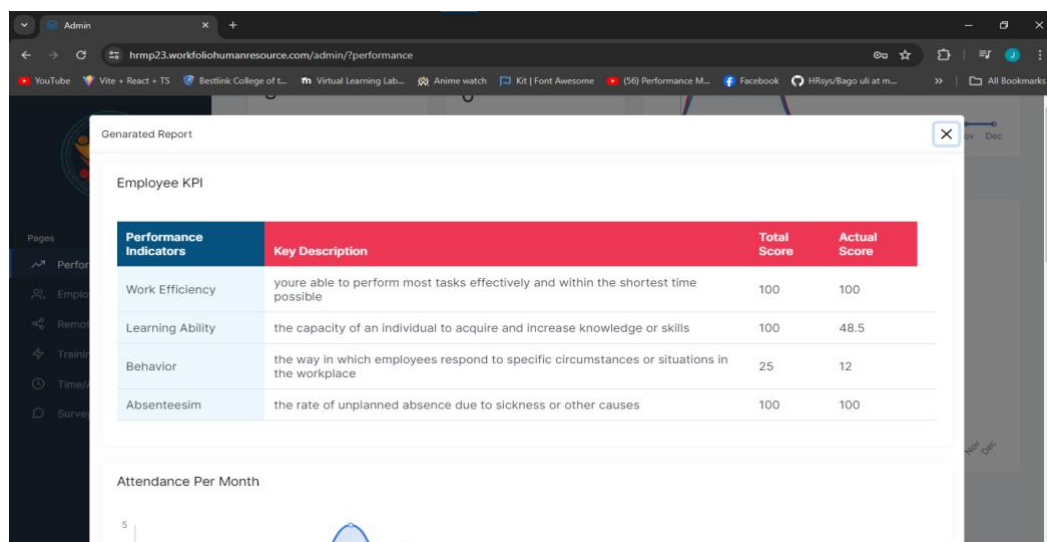


Figure 25: Generated Report

- **Employee Attendance Per Month**
- **Employee Completed Tasks**

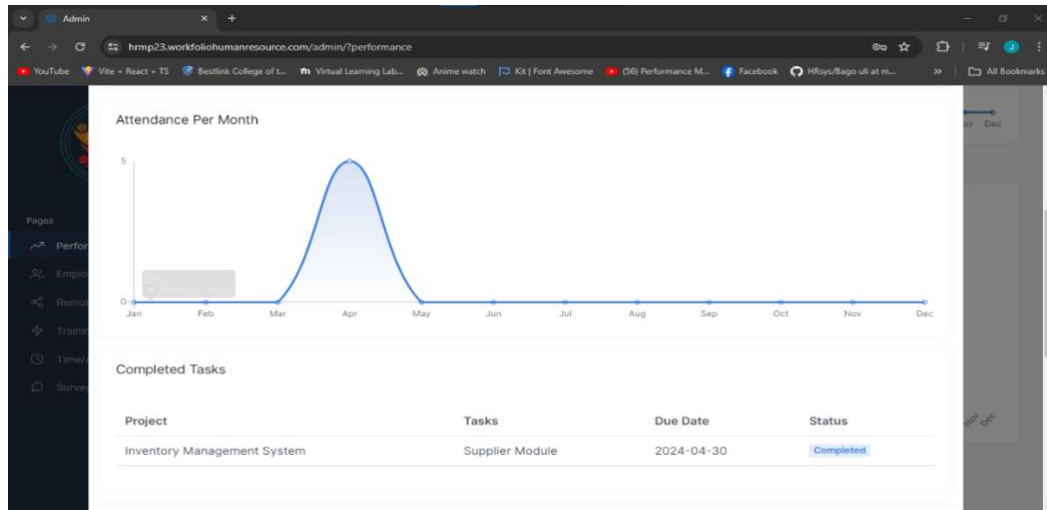
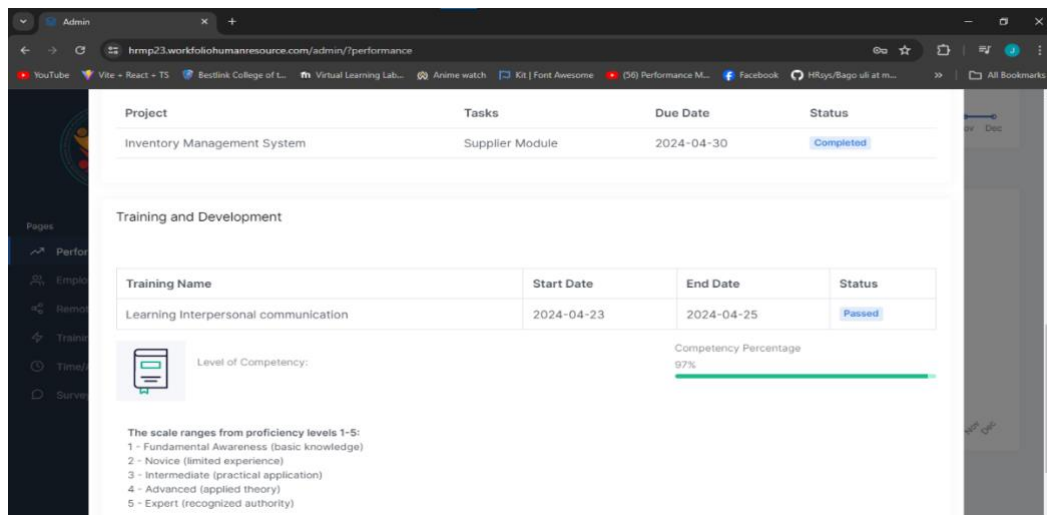


Figure 26: Attendance Per Month

- **Employee Training and Development and his Competency Percentage**
- **Scroll down** the screen you will see the Print Report button to print the KPI of the Employee



- Click **Print Report**

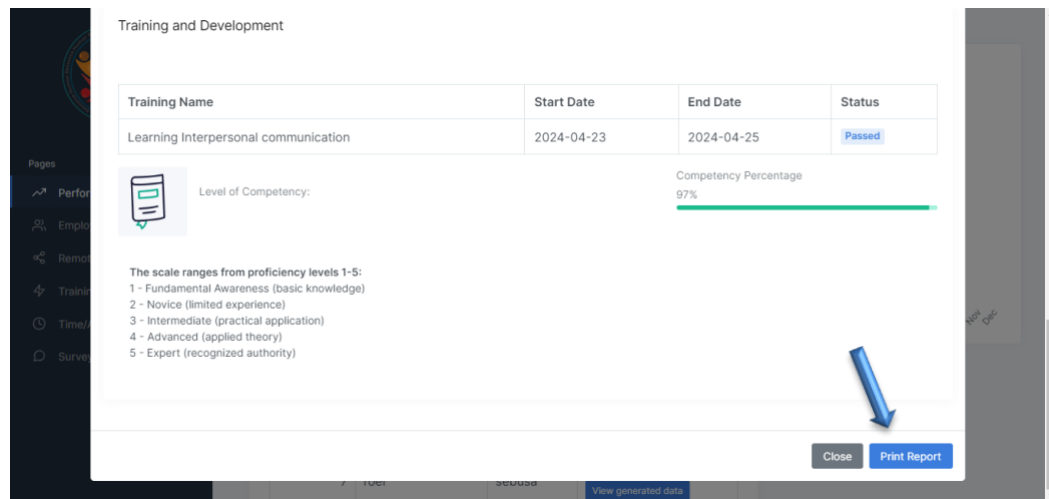


Figure 27: Task & Print

- After Clicking **Print Report** you will see the full KPI of the Employee.
- Click **“Print”**.

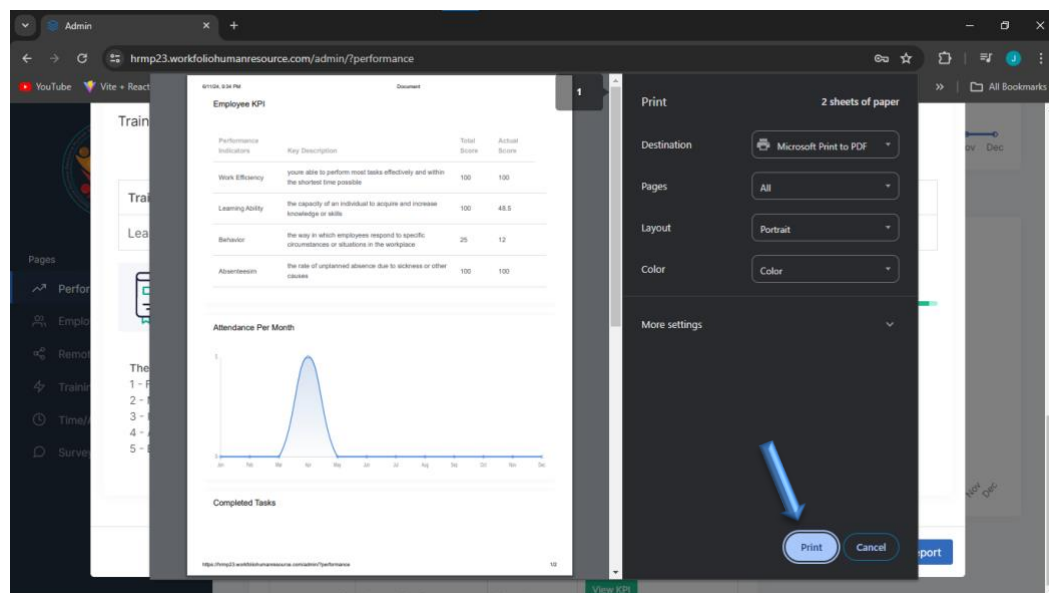


Figure 28: Task Details

- To see the Employee's Generated Data, click **“View Generated Data”**.
- After Clicking you will see the percentages of his Performance.

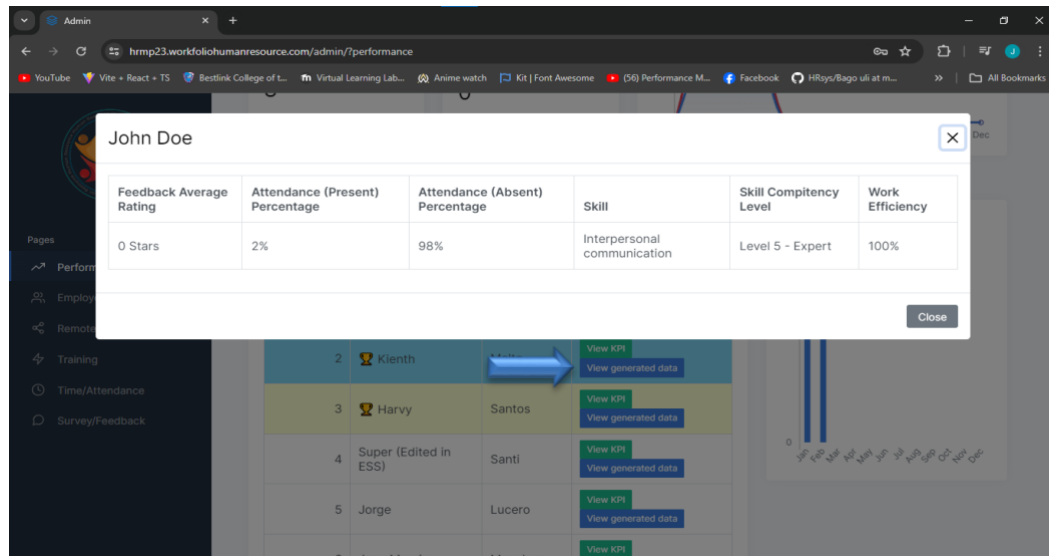


Figure 29: Employee Status

- Click the sidebar to move to other sub-modules **“Click Employee”** to see the employee list.

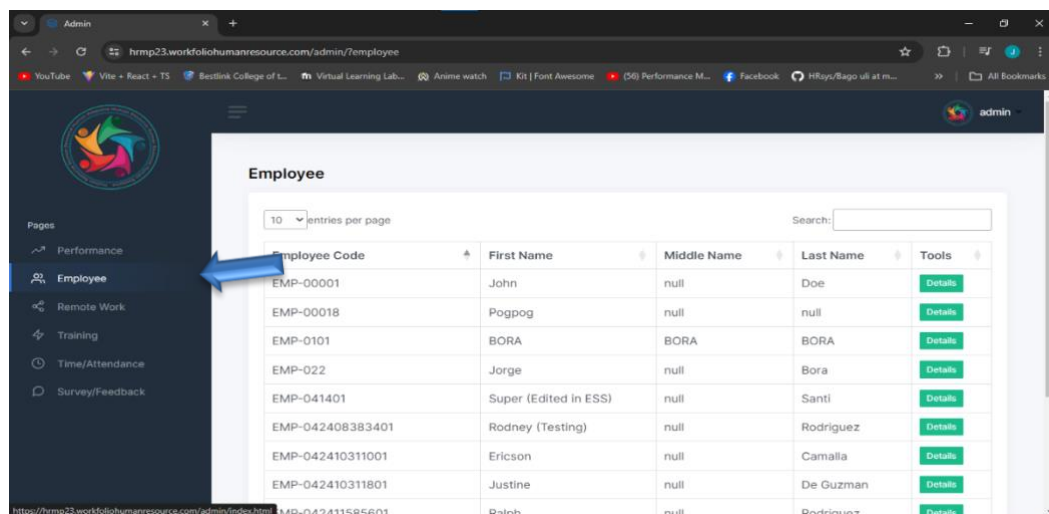
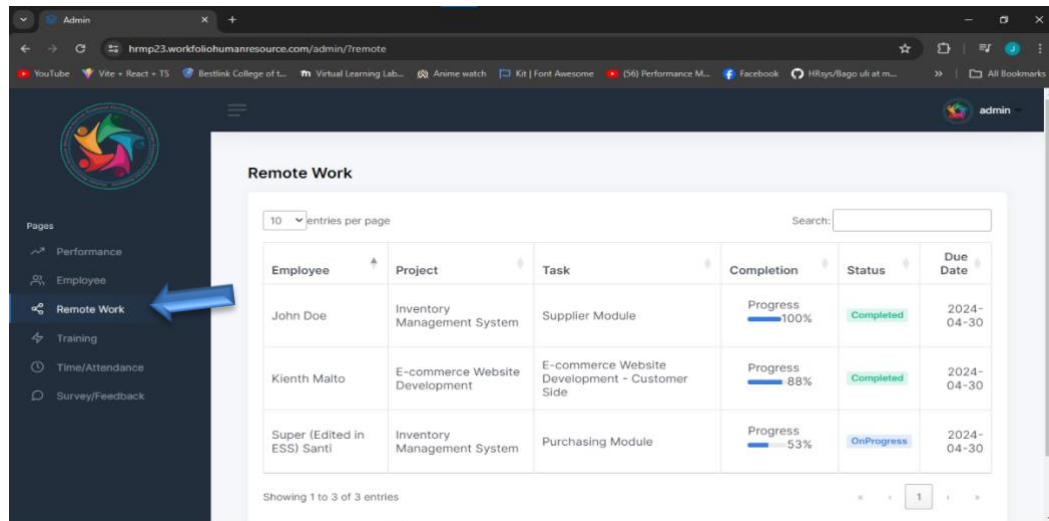


Figure 30: Sub Module 1: Employee Details

- Click the sidebar to move to other sub-modules “**Click Remote Work**” to see the employees in the Remote Work.



Remote Work

10 entries per page

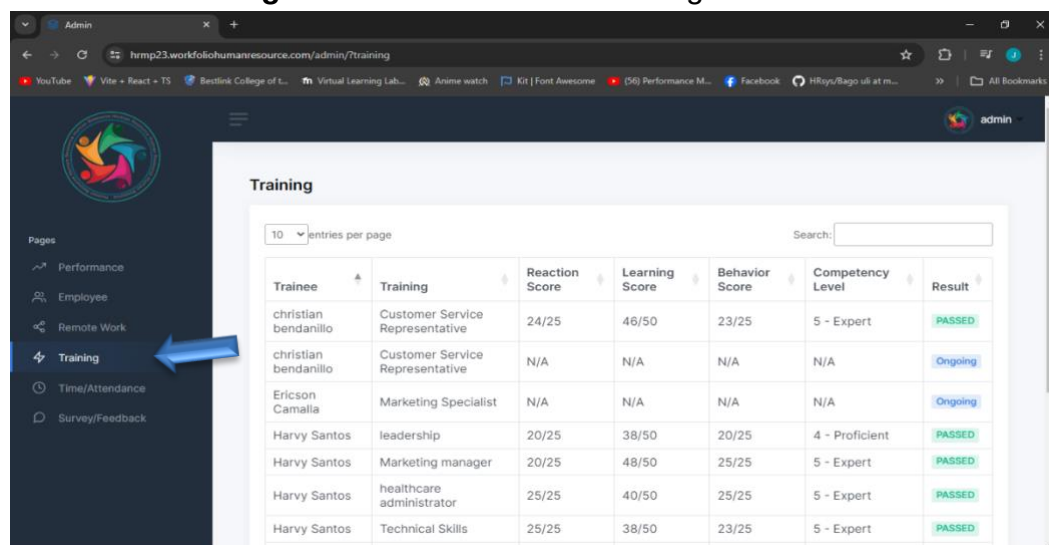
Search:

Employee	Project	Task	Completion	Status	Due Date
John Doe	Inventory Management System	Supplier Module	Progress 100%	Completed	2024-04-30
Kienth Maito	E-commerce Website Development	E-commerce Website Development - Customer Side	Progress 88%	Completed	2024-04-30
Super (Edited in ESS) Santi	Inventory Management System	Purchasing Module	Progress 53%	OnProgress	2024-04-30

Showing 1 to 3 of 3 entries

Figure 31: Sub Module 2: Remote Work

- Click the sidebar to move to other sub-modules “**Click Training**” to see who is in the Training and their Results.



Training

10 entries per page

Search:

Trainee	Training	Reaction Score	Learning Score	Behavior Score	Competency Level	Result
christian bendanillo	Customer Service Representative	24/25	46/50	23/25	5 - Expert	PASSED
christian bendanillo	Customer Service Representative	N/A	N/A	N/A	N/A	Ongoing
Ericson Camalla	Marketing Specialist	N/A	N/A	N/A	N/A	Ongoing
Harvy Santos	leadership	20/25	38/50	20/25	4 - Proficient	PASSED
Harvy Santos	Marketing manager	20/25	48/50	25/25	5 - Expert	PASSED
Harvy Santos	healthcare administrator	25/25	40/50	25/25	5 - Expert	PASSED
Harvy Santos	Technical Skills	25/25	38/50	23/25	5 - Expert	PASSED

Figure 32: Sub Module 3: Training

- Click the sidebar to move to other sub-modules “**Click Time and Attendance**” to see the employee’s Attendance.
- Click the “**View**” to see the Attendance Average.

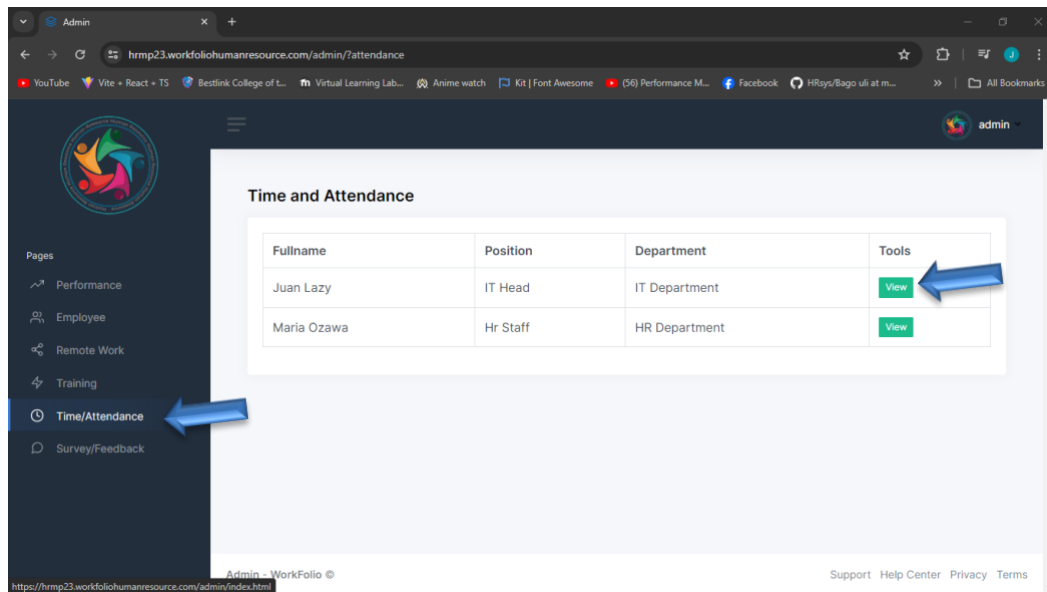


Figure 33: Sub Module 4: Time and Attendance

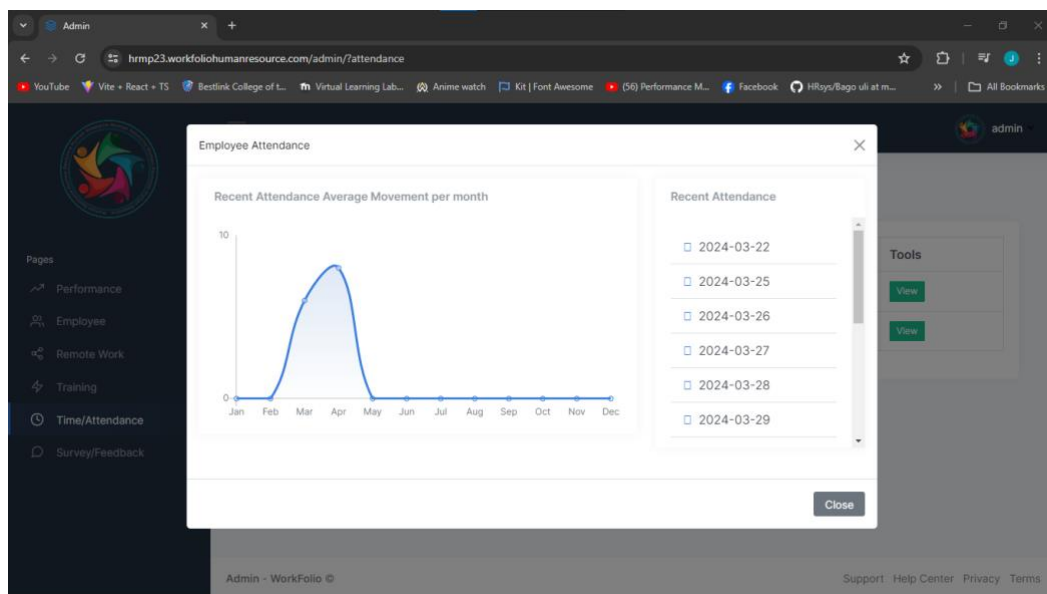


Figure 34: Sub Module 4: Time and Attendance Details

- Click the sidebar to move to other sub-modules “**Click Survey/Feedback**” to see the employee list feedback.

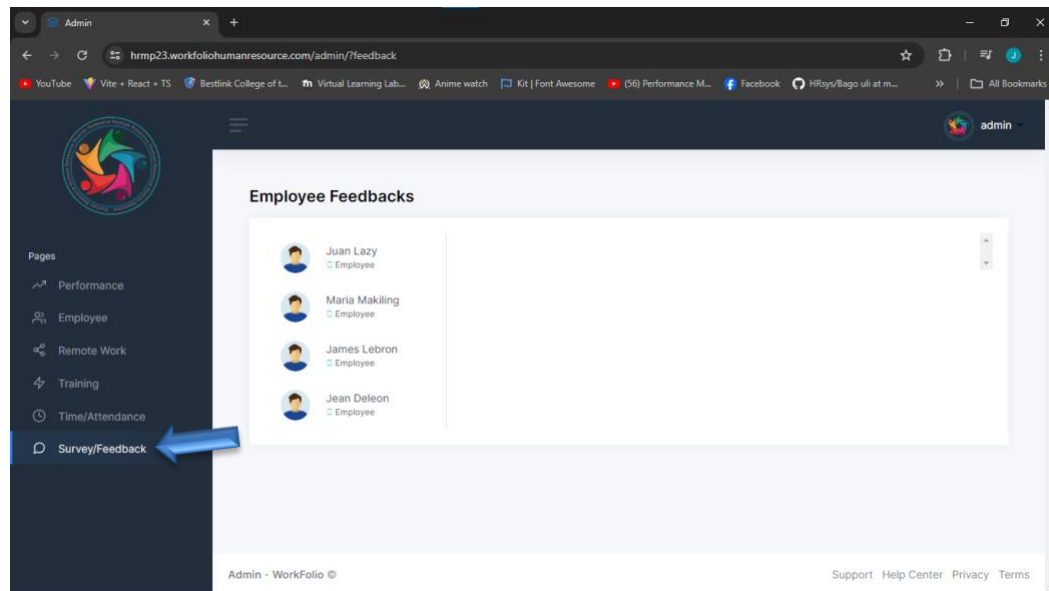


Figure 35: Sub Module 5: Survey and Feedback

16.12 Known Issues and Troubleshooting

List of Known Issues and Their Status

Issue: Unable to Save Changes

Status: Under investigation.

Issue: deleting documents was not functional

Status: Being addressed in the next software update

Troubleshooting Steps:

1. Check physical connections
 2. Restart your devices
 3. Refresh the page and try saving the invoice again.
 4. Contact your ISP
- **Troubleshooting Steps:**
 1. Confirm that the email recipient has received the message by checking their inbox, which includes examining spam or junk folders for any potential misplacement of the email.
 2. Ensures that the correct email address is linked to the account.
 3. If the problem persists, contact technical support for help.

Contact Information for Technical Support

For technical assistance and further inquiries, please reach out to our support team:

Name	Email	Phone
Palomo, Angela Nicole L.	palomoangelanicole@gmail.com	63951 927 9593
Francisco, Johnlester E.	johnlesterfrancisco3@gmail.com	639919464906
Sedano, Justine Alen A.	alensedano756@gmail.com	639063103249
Pajulas, Arvin-Jay O.	pajulasarvinjay@gmail.com	639511444579
Saguisa, Dennis E.	dsaguisa09@gmail.com	639683262078

Table 8: Technical Support

Repository location and access details

<https://workfoliohumanresource.com/>

<https://hrmp23.workfoliohumanresource.com/admin/?performance>

Network Latency: Assess the impact of network speed on repository operations, particularly for distributed teams.

16.13 Version Control and Source Code Repository

➤ Commit and Pull Times:

Measure how long it takes to commit changes and pull updates from the repository.

➤ Merge and Build Times:

Track the time required to merge branches and build the project.

➤ Repository Size and Growth:

Monitor repository size and the rate at which it grows.

Optimization Techniques

- **Repository Structure:** Maintain a clear and modular repository structure to simplify navigation and operations.
- **Archiving Old Data:** Archive or delete obsolete branches and files to reduce repository size.

- Garbage Collection: Regularly perform garbage collection and repository cleanup using tools like `git gc` for Git.
- Sharding and Submodules: Break large repositories into smaller ones or use submodules to manage dependencies and reduce complexity

. Branching Strategies

Common Branching Models

- Feature Branching:

Each feature or bug fix is developed in its own branch.

- Release Branching:

Branches are created for each release to stabilize the code before deployment.

- Hotfix Branching:

For quick fixes, branches are created off the main branch to address critical issues.

- GitFlow:

A comprehensive model with dedicated branches for features, releases, and hotfixes.

- Trunk-Based Development:

Developers commit to a single trunk branch, with frequent integrations to ensure stability.

Merging Strategies

Common Merging Techniques

- Fast-Forward Merges: Simple merges where the branch is directly moved forward to include new commits.
- Three-Way Merges: Used when branches diverge, involving a common ancestor and two branch tips.
- Rebase and Merge: Replays commits from one branch onto another, creating a linear history.
- Squash and Merge: Combines all commits from a branch into a single commit before merging.

16.14 Regarding CI/CD Pipeline Efficiency

- Constant Feedback:

To promote speedy fixes, give developers immediate feedback on build and test results.

- Automated notifications:

To guarantee prompt interventions, set up notifications for build failures or performance regressions.

- Regular Retrospectives:

Review pipeline performance and pinpoint opportunities for development by doing retrospectives on a regular basis.

- Regarding Worker Performance

- Training and Development:

To close skill gaps and boost productivity, provide training courses based on analytics.

- **Acknowledgment and Rewards:**

Give credit for contributions made to the development process as a whole as well as the CI/CD pipeline.

- **Balanced Workload:**

To guarantee a fair workload distribution and prevent overburdening certain team members, employ analytics.

Automation scripts and tools used

Testing Automation Scripts

JUnit/NUnit: For unit testing in Java and .NET applications, respectively.

Selenium/Cypress: For automating browser-based tests, ensuring web applications work as expected.

Postman/Newman: For API testing automation. Postman allows for creating tests, and Newman runs them in a CI environment.

Deployment Automation Scripts

Ansible Playbooks: For automating deployment tasks, configuration management, and provisioning servers.

Terraform Scripts: For infrastructure as code, managing cloud resources such as AWS, Azure, and GCP.

Docker Compose/YAML Files: For defining multi-container Docker applications and deploying them.

Deployment strategies and rollback procedures

Blue-Green Deployment

Description: This strategy involves two identical environments: Blue current live environment and Green new environment for deployment. Traffic is routed to Blue, and once Green is verified, traffic is switched to Green.

Benefits: Minimal downtime, quick rollback by switching back to Blue, easy testing in production-like environment.

Use Cases: Ideal for applications requiring zero downtime and smooth user experiences.

Immediate Rollback

Description: Quickly revert to the previous stable version in case of critical issues.

Steps:

Trigger Rollback: Use deployment tool to revert to the previous version, reset, rollback command in Kubernetes.

Monitor: Ensure the rollback is successful and the application is stable.

Post-Mortem: Analyze the issue to prevent future occurrences.

16.15. Licensing and Open-Source Libraries:

Information about licenses for software and libraries used

Complying with and using PHP and its libraries properly requires an understanding of the licenses that govern them. PHP can be freely used, modified, and distributed under the terms of the PHP License Version 3.01, provided that copyright notices are kept in place and that using the term "PHP" without authorization is prohibited. PHP libraries are commonly licensed under the permissive MIT License, which permits unlimited use and necessitates citing the license in copies; or the GNU General Public License (GPL), which requires derivative works to be made available under the same license.

It is necessary to comprehend the licenses of the packages you include in order to use Composer to manage dependencies for PHP projects. The most popular licenses, each with their own set of rules and regulations, are the MIT, GPL, LGPL, Apache License 2.0, and BSD. Maintaining

compliance entails following distribution and modification guidelines, especially for copyleft licenses, giving due acknowledgment to the original authors, ensuring compatibility, and including the original license wording in distributions.

Credits and attributions for open-source components

Read the License: Before using any open-source component, make sure you have read and comprehended its license in its entirety. Additional attribution requirements may apply to different licenses. Examine the README or documentation files for the open-source components. Writers frequently specify exactly how they wish to be attributed. Include the open-source component license files in your project. This aids users in understanding the conditions under which the program is distributed and is frequently a prerequisite of many open-source licenses. Include a section in your project called "About" or "Credits" where you can mention all the open-source components along with the creators of each one.

Give concise, precise details on each component. For every open-source component, please include the element's name, the names of the author(s) or contributors, and the license(s) under which it is distributed. **Link to Project Pages:** If possible, provide links to the websites, repositories, or project pages of the open-source components. This provides due credit and enables users to get more component information.

Respect License Guidelines: Certain licenses may specify exactly how credit must be given. As an illustration, the MIT license frequently demands that the original copyright notice be included. Follow these instructions to guarantee adherence. Handle Dependencies: If the dependencies in your project are handled by package managers such as npm (for JavaScript) or composer (for PHP), you should go to the documentation of each package management to see how attributions are supposed to be made. Respect License Guidelines: Certain licenses may specify exactly how credit must be given. As an illustration, the MIT license frequently demands that the original copyright notice be included.

Follow these instructions to guarantee adherence. Handle Dependencies: If the dependencies in your project are handled by package managers such as npm (for JavaScript) or composer (for PHP), you should go to the documentation of each package management to see how attributions are supposed to be made. Be open and honest about the open-source parts you utilize. This data adds to the spirit of the open-source community and is beneficial to both developers and users. Make a note of the versions, attributions, and open-source components that were used. Maintaining compliance and conducting audits may benefit from this.

16.16. Performance Metrics and Monitoring:

Metrics collected and monitored

- Response time:

Analyze how long it takes the system to react to user requests to get a sense of how responsive the program is overall.

- Throughput:

Keep an eye on how quickly the system responds to queries to determine its efficiency and capacity.

- Error rates:

Monitor the percentage of errors or unsuccessful requests to find problems with system stability and user experience.

- Resource Utilization

To guarantee optimal resource utilization and identify possible bottlenecks, track CPU, memory, and disk usage.

- Network Latency

Analyze the time it takes for data to move between various components to help find performance problems with the network.

- Database Performance

Monitor database-specific data, including connection pool utilization, indexing efficiency, and query execution time.

- HTTP Source Codes

Watch HTTP status codes for indications of problems with communication between the client and the server.

- Concurrency

Keep an eye on the number of active users or connections to gauge how well the system can manage several requests at once.

Tools and dashboards used for performance analysis

A complete performance benchmarking tool made especially for PHP web applications is called Blackfire. It provides thorough insights into a number of application performance-related topics, including as memory utilization, database queries, function call traces, and execution durations. Developers can identify performance bottlenecks and effectively adjust their code to increase the responsiveness and scalability of their applications by examining this data.

Grafana is a flexible analytics and monitoring dashboard that facilitates the visualization of data from a range of sources, such as time-series databases, logs, and performance indicators. Grafana enables users to build customized dashboards with interactive graphs, charts, and alerts that facilitate real-time application performance analysis and monitoring. It is a well-liked option for displaying and evaluating performance data in a

variety of contexts, including PHP online applications, due to its adaptability and extensive integration capabilities.

Actions taken based on performance data

✓ Automatic Scaling:

Use auto-scaling techniques to dynamically modify resources in response to demand, guaranteeing peak performance even during periods of high traffic.

✓ Alerts and Notifications:

To get instant notifications when performance metrics depart from acceptable limits, set up alerts for important thresholds.

✓ Capacity Planning:

Utilize capacity planning to make proactive resource scaling decisions and account for future expansion by leveraging historical performance data.

✓ Code Optimization:

To find areas for code optimization, enhancements, and improvements, analyze performance measurements.

✓ Infrastructure Tuning:

Utilizing performance data, modify infrastructure configurations to maximize resource allocation.

✓ Database Indexing and Query Optimization:

By identifying inefficient queries, building effective indexes, and fine-tuning database parameters, you may maximize database performance.

✓ Cache Optimization:

To speed up response times, adjust caching tactics based on cache hit rates and performance indicators.

✓ Incident Response:

To promptly address and fix issues, create incident response strategies based on performance monitoring data.

✓ Continuous Improvement:

Performance data can be used to guide a plan for continuous improvement, which will improve system efficiency and help to refine the application architecture.

In conclusion, our all-encompassing performance monitoring approach include gathering a variety of measurements, applying strong analytical tools, and initiating preemptive measures to maximize system performance, guarantee a flawless user experience, and preserve the integrity of the entire infrastructure.